

PROJECTIVE INFINITIES AND b -CALCULUS

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ABSTRACT. For a manifold $\overline{M}$ with boundary ∂M and interior M , we introduce and study a weakening of the concept of projective compactness for torsion-free linear connections on M , which we call projective pre-compactness. Via the Levi-Civita connection, this concept applies to pseudo-Riemannian metrics on M . This is motivated by applications to scattering theory and, especially, to general relativity (GR), in view of asymptotic forms of metrics used in these areas.

In the general setting of a projectively pre-compact connection ∇ we show that, assuming weak conditions on the asymptotics of Ricci curvature, there is an induced projective structure on the boundary. Under a slightly stronger condition on the Ricci curvature, we show explicitly that the standard tractor bundle and its normal tractor connection arise naturally on this boundary structure. The key ingredient to this is that ∇ admits a smooth extension to the boundary as a linear connection on the tensor product of Melrose's b -tangent bundle with an appropriate density bundle, which then restricts to the boundary tractor bundle.

A projectively pre-compact pseudo-Riemannian metric (satisfying the conditions on the asymptotics of Ricci curvature) is then shown to induce a holonomy reduction of the boundary projective structure to an indefinite orthogonal group. This endows the boundary with a decomposition into so-called curved orbits, which are either open or embedded hypersurfaces, representing space-like, time-like and light-like infinities in a GR context. We introduce and study a new asymptotic form for such metrics which is available near any boundary point and relate it to an asymptotic form used in general relativity. The latter is only available near boundary points in the open curved orbits. We show that, in that region, projective pre-compactness essentially is equivalent to this asymptotic form from GR, and that projective compactness is equivalent to the vanishing of the mass aspect function.

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1. INTRODUCTION

Conformal compactification was originally introduced by R. Penrose in the setting of general relativity, and hence of Lorentzian metrics, with the aim to attach a “boundary at infinity” to non-compact space-times, and remains extremely important [13, 15, 22]. Formally, this means finding a conformal embedding of the space-time into a compact pseudo-Riemannian manifold and then considering the (compact) closure of the image. A major motivation for the concept is that in pseudo-Riemannian geometry null geodesics are conformally invariant up to parametrization, so one can study “where light rays meet infinity” in this picture. The basic concept makes sense in any signature and in the nicest cases this leads to a conformal embedding of the space as the interior of a pseudo-Riemannian manifold with boundary that has a specific behavior of the metric towards infinity. In particular, this happens in many cases for complete pseudo-Riemannian manifolds which are asymptotically hyperbolic, in an appropriate sense, and the metric then induces a conformal structure on the boundary (i.e., the “conformal infinity”). Requiring the interior to be (Riemannian and negative) Einstein, one arrives at the concept of Poincaré–Einstein manifolds which have become a major research topic in geometric analysis over the last decades [1, 14, 17, 18, 20, 19].

Already the example of Minkowski space shows that things are complicated in an asymptotically flat setting. Here the “boundary” is a union of hypersurface components which represent light-like infinity and points glued to these, which represent space-like and time-like infinities. While the behavior around light-like infinities of a conformal compactification in this case is very satisfactory, dealing with space-like and time-like infinities is subtle both from the GR point of view and mathematically. In the GR literature attempts to find other compactifications at space-like and time-like infinity were made early, see e.g. [3] and in particular [16], where it is indicated that such an alternative compactification can be “almost” compatible with the projective structure underlying the pseudo-Riemannian metric. In these approaches, one obtains a boundary at infinity of hypersurface type and there is the natural question of whether this inherits a projective structure, making it into a *projective infinity*.

The projective structure induced by a pseudo-Riemannian metric can be viewed as the family of all *geodesic paths* (or unparametrized geodesics) of the Levi-Civita connection. More formally, one can view it as the equivalence class of all torsion-free linear connections on the tangent bundle, which have the same geodesics as the Levi-Civita connection up to parametrization. In the GR literature, these geometric aspects remain in the background, the focus is on a certain asymptotic form for the pseudo-Riemannian metrics, in such a compactification, near space-like and time-like infinity [3, 2]. A special case of this asymptotic form is studied as a “scattering metric” in geometric scattering theory, see Chapter 6 of [21].

One possibility to obtain a projective infinity comes from a general theory of projective compactifications, which was developed in [9, 10] in the setting of torsion-free linear connections on the interior of a manifold with boundary. A change

from a such a linear connection to a projectively related one is described by a one-form. The basic definition of projective compactness is that using a local defining function for the boundary to construct a one-form (that does not admit a smooth extension to the boundary), the resulting projective modification of the given connection *does* admit such an extension. So similarly to conformal compactness, where one requires the metric to diverge in a specific way towards the boundary, one here requires the connection to diverge in a specific way. In particular, since projective modifications of the given connection admit a smooth extension to the boundary, the projective structure determined by the connection admits a smooth extension.

A new feature of projective compactness is that the concept allows for and requires an additional parameter β , called the *order of projective compactness*, which turns out to be related to volume growth. This is a positive real number, assumed to be ≤ 2 to ensure that the boundary is at infinity, see [9] for details. Depending on β , a projective compactification may induce different structures on the boundary, the main cases interest (so far) are $\beta = 1$ and $\beta = 2$. For $\beta = 2$, one obtains a conformal structure on the boundary and the resulting theory is rather closely related to (but different from) conformal compactifications. The relevant case for the current article is $\beta = 1$. Assuming that the boundary is totally geodesic, which amounts to asymptotic conditions on the Ricci curvature, one here obtains an induced projective structure on the boundary.

In particular, one may apply this to the flat connection on $\mathbb{R}^{n+1}$, for which, via central projection, the compactification can be viewed as a closed hemisphere in S^{n+1} with its standard (flat) projective structure and the equator as the boundary at infinity. Adding the Minkowski metric into the picture gives rise to an additional structure on the boundary, which can be formalized as a holonomy reduction in the sense of [12] of the canonical Cartan connection, associated to the projective structure on the boundary at infinity, to the group $SO(n, 1) \subset SL(n+1, \mathbb{R})$. This in particular decomposes the boundary sphere into five so-called *curved orbits*. Two of these are isomorphic copies of n -dimensional hyperbolic space, one is a copy of n -dimensional de-Sitter space, and a further two are copies of a conformal Riemannian sphere S^{n-1} .

One of the contributions of [9] is extending this picture to general Ricci-flat connections and pseudo-Riemannian metrics, which are projectively compact of order 1. In the case of metrics, the work [9] also studies the relation to asymptotic forms

$$g = \frac{C}{\rho^4} d\rho^2 + \frac{h}{\rho^2}$$

for a local defining function ρ for the boundary. Here C is a smooth function that is asymptotic to ± 1 to second order along the boundary, i.e. such that $C = \pm 1 + \rho^2 f$ for a function f that admits a smooth extension to the boundary. On the one hand, it is shown in [9] that any metric that admits such a form is projectively compact of order 1. For Ricci-flat metrics, [9] also proves a converse to this result, namely that such an asymptotic form is always available (for appropriate defining functions) locally around points of the open curved orbits from above. All the developments

in [9] heavily rely on tractor calculus for projective structures, which provides an equivalent encoding of the canonical Cartan connection and emphasises the analogy to conformal geometry.

While these are very satisfactory results, there is a significant drawback from the point of view of general relativity. In the asymptotic forms considered in the physics literature, see e.g. [3] and [2], one requires that the function C mentioned above has the form $\pm 1 + \rho f$ and the boundary value of f (whose vanishing is equivalent to the form required for projective compactness), called the *mass aspect*, is closely related to the total mass of the space-time. Thus true projective compactness seems to be incompatible with non-zero mass. This was also emphasized in the recent articles [5, 6, 7], which explicitly use the concept of projective compactness from [9]. However, it was noted in these works that for the more general asymptotic form one obtains most of the features of projective compactness.

The current article addresses and clarifies exactly this issue. We introduce a weakening of the concept of projective compactness that we call *projective pre-compactness*. We consider the same projective modification as in the definition of projective compactness, but do not require that the resulting connection extends smoothly to the boundary. Rather we define and require a very slight weakening of this condition. Of course, this means that the projective structure determined by a projectively pre-compact connection does *not* (in general) admit a smooth extension to the boundary. (This is required. Indeed, as shown in [9] a Ricci flat connection is necessarily projectively compact of order one if its projective structure extends.) Hence the basic tools used for the study of projective compactness so far are not available in our setting. Nevertheless we are able to prove that the main features of projectively compact connections and metrics are still available in this more general setting. Even in the projectively compact case, the results we obtain are stronger than the existing ones while proofs are significantly simpler in general.

The main technical tool we use to replace tractors is the *b-tangent bundle* introduced by R. Melrose and used in his school of geometric analysis (see e.g. [21]). Given a manifold $\overline{M}$ with boundary ∂M and interior M , this is a vector bundle $T^b\overline{M} \rightarrow \overline{M}$, whose sheaf of smooth sections is the sheaf $\mathfrak{X}_\partial$ of vector fields on $\overline{M}$ which are tangent to ∂M along ∂M . The existence of such a bundle follows from the fact that $\mathfrak{X}_\partial$ is a locally free sheaf. Once this vector bundle is available, one can of course apply all the tools from differential geometry to it, and particular the notion of a linear connection makes sense. Our key technical result is that a projectively pre-compact linear connection on the interior M of $\overline{M}$ extends to a linear connection on $T^b\overline{M}(-1)$, the tensor product of $T^b\overline{M}$ with an appropriate density bundle. While we don't use tractors at this stage, it should be emphasized that the idea for this approach comes from tractors, see Section 2.10 for details.

An overview of the results we obtain is as follows. Section 2 studies projective pre-compactness in the setting of connections. After introducing the concept, we prove that, for $\overline{M} = M \cup \partial M$, a projectively pre-compact linear connection ∇ on M gives rise to a canonical weighted symmetric bilinear form Φ on $T\overline{M}|_{\partial M} \times T\partial M$, which restricts to a weighted second fundamental form on $T\partial M \times T\partial M$,

see Theorem 2.5. This form is closely related to the asymptotics of the Ricci curvature of ∇ , see Theorem 2.9. The vanishing of this second fundamental form characterises a totally geodesic boundary and, assuming this, we show in Theorem 2.9 that ∇ induces a projective structure on the boundary ∂M , so this becomes the *projective infinity* of ∇ .

For the next step, we need a slightly stronger assumption, namely that the canonical weighted form Φ on $T\overline{M}|_{\partial M} \times T\partial M$ vanishes identically. This can also be characterized in terms of the asymptotics of the Ricci curvature (or, equivalently, the Schouten tensor) of ∇ and we refer to it as the boundary being *strongly totally geodesic*. Under this assumption, Theorem 2.12 shows that ∇ extends to a linear connection on the bundle $T^b\overline{M}(-1)$ which, in spite of the simple proof, is the main technical input for all further developments. Our main result in the general setting of connections is Theorem 2.13, which shows that the standard tractor bundle and tractor connection associated to the induced projective structure on the boundary arise as restrictions from the weighted b -tangent bundle and its induced connection.

Theorem. *For $\overline{M} = M \cup \partial M$, let ∇ be an affine connection on M , which is projectively pre-compact and let P be the Schouten tensor of ∇ . Suppose further that for all $\xi, \eta \in \mathfrak{X}(\overline{M})$, the function $P(\xi, \eta)$ admits a smooth extension to the boundary if at least one of the fields lies in $\mathfrak{X}_\partial(\overline{M})$ and its boundary value vanishes if both fields lie in $\mathfrak{X}_\partial(\overline{M})$. Then the bundle $T^b\overline{M}(-1)|_{\partial M}$ with the connection induced by ∇ can be naturally identified with the projective standard tractor bundle of the induced projective structure on ∂M and its normal tractor connection.*

In Section 3, we study projective pre-compactness of Levi-Civita connections of pseudo-Riemannian metrics. We start by considering metrics with an asymptotic form

$$(3.2) \quad g = \frac{h}{\rho^2} + \frac{C}{\rho^4} d\rho^2,$$

where $C = \pm 1 + \rho f$ for $f \in C^\infty(\overline{M}, \mathbb{R})$ and $h \in \Gamma(S^2 T^* \overline{M})$ is non-degenerate along the boundary on $T\partial M \times T\partial M$. Following ideas in [9], we prove in Theorem 3.1 that g (i.e. its Levi-Civita connection) is projectively pre-compact and it is projectively compact if $f|_{\partial M} \equiv 0$.

Conversely, we next assume that g is a projectively pre-compact pseudo-Riemannian metric, which is asymptotically tangentially Ricci flat in the sense that $\text{Ric}(\xi, \eta)$ admits a smooth extension to the boundary if one of its entries lies in $\mathfrak{X}_\partial(\overline{M})$ and vanishes along the boundary if both of its entries have this property. Under this assumption, we show that there is an induced projective structure on the boundary, and we prove in Theorem 3.3 that, similarly as in the projectively compact case (but even in that case under weaker assumptions), g gives rise to a reduction of projective holonomy of the boundary structure to an indefinite orthogonal group. By general results on such reductions, this decomposes $\partial M = \partial M_+ \sqcup \partial M_0 \sqcup \partial M_-$ into so-called curved orbits, where $\partial M_\pm$ are open and ∂M_0 (if non-empty) is an embedded closed hypersurface. This is the decomposition of the boundary into time-like, light-like and space-like infinity. It is easy to see

that an asymptotic form (3.2) can only be available locally around points in $\partial M_{\pm}$ and for special defining functions. Our second main result in Theorem 3.5 provides an asymptotic form that is available everywhere together with strong additional information on the coefficients in the asymptotic form. In this result, ∇^{ρ} denotes the linear connection on $T\partial M$ induced by a specific projective modification of ∇ determined by the defining function ρ and $(\lambda \odot d\rho)(\xi, \eta) := \frac{1}{2}(\lambda(\xi)d\rho(\eta) + \lambda(\eta)d\rho(\xi))$.

Theorem. *For $\overline{M} = M \cup \partial M$, let g be a pseudo-Riemannian metric of signature (p, q) on M which is projectively pre-compact and asymptotically tangentially Ricci flat, and let ρ be any defining function for ∂M .*

(1) There are a smooth function $C : \overline{M} \rightarrow \mathbb{R}$, a one-form $\lambda \in \Omega^1(\overline{M})$ and a section $h \in \Gamma(S^2 T^ \overline{M})$ such that, over M , we have*

$$(3.6) \quad g = \frac{Cd\rho^2}{\rho^4} + \frac{\lambda \odot d\rho}{\rho^3} + \frac{h}{\rho^2}.$$

(2) Given any asymptotic form as in (3.6), let $\underline{C}$ be the boundary value of C , and let $\underline{\lambda} \in \Omega^1(\partial M)$ and $\underline{h} \in \Gamma(S^2 T^ \partial M)$ the restrictions of the boundary values of λ and h to tangential entries. Then $\underline{C}$ is a defining function for ∂M_0 and the symmetric bilinear form on the bundle $T\partial M \oplus \mathbb{R}$, over ∂M , defined by*

$$((\xi, a), (\eta, b)) \mapsto \underline{C}ab - \frac{1}{2}a\underline{\lambda}(\eta) - \frac{1}{2}b\underline{\lambda}(\xi) + \underline{h}(\xi, \eta)$$

is non degenerate of signature (p, q) . Moreover, denoting by ∇^{ρ} the induced connection on $T\partial M$ and by P^{ρ} its Schouten tensor, we get for $\xi, \eta, \zeta \in \mathfrak{X}(\partial M)$

$$\begin{aligned} \underline{\lambda} &= -dC & \underline{h}(\xi, \eta) &= -\frac{1}{2}(\nabla_{\xi}^{\rho} \underline{\lambda})(\eta) + \underline{C}P^{\rho}(\xi, \eta) \\ (\nabla_{\xi}^{\rho} \underline{h})(\eta, \zeta) &= \frac{1}{2}(P^{\rho}(\xi, \eta)\underline{\lambda}(\zeta) + P^{\rho}(\xi, \zeta)\underline{\lambda}(\eta)). \end{aligned}$$

This easily leads to a proof of existence of an asymptotic form (3.2) locally around points at which $C \neq 0$, see Corollary 3.7.

We conclude the article by proving characterizations of projective compactness in terms of the asymptotic forms (3.2) and (3.6). For (3.6), Theorem 3.8 provides an equivalent condition for projective compactness of g in terms of asymptotic differential equations on the coefficients C and λ in (3.6). In the case of (3.2), it remains to show that for $C = \pm 1 + \rho f$, projective compactness of g implies that $f|_{\partial M} \equiv 0$. This is done in Theorem 3.9 under the slightly stronger curvature assumption that the Ricci curvature of g admits a smooth extension to the boundary and its boundary value vanishes if one of its entries is tangent to the boundary. The proof shows that this curvature assumption, which seems to be always satisfied in GR applications, is necessary.

2. PROJECTIVE PRE-COMPACTNESS FOR CONNECTIONS

In this Section, we define and study the concept of projective pre-compactness in the general setting of torsion-free connections. The main focus is on the existence of an induced projective structure on the boundary and an explicit description of the projective standard tractor bundle determined by this structure.

2.1. Setup and background. We consider a smooth manifold $\overline{M}$ with boundary ∂M and interior M . A local defining function for ∂M is then a smooth, real valued function ρ defined on an open subset $U \subset \overline{M}$ such that $\rho^{-1}(\{0\}) = U \cap \partial M$ and such that $d\rho|_{U \cap \partial M}$ is nowhere vanishing. A key feature of this concept is that a smooth function $f : U \rightarrow \mathbb{R}$ vanishes on $U \cap \partial M$ if and only if it can be written as $f = \rho f_1$ for some smooth function $f_1 : U \rightarrow \mathbb{R}$. In this case we say that $f = \mathcal{O}(\rho)$. Similarly, for $k \geq 2$, we say that f vanishes to k th order along ∂M or that $f = \mathcal{O}(\rho^k)$ if and only if $f = \rho^k f_1$ for a smooth function $f_1 : U \rightarrow \mathbb{R}$.

For a function $f : U \cap M \rightarrow \mathbb{R}$, we will say that $f = \mathcal{O}(\rho^{-1})$, if the smooth function $\rho f : U \cap M \rightarrow \mathbb{R}$ admits a (necessarily unique) smooth extension to all of U . Similarly for $k \geq 2$, we say that $f = \mathcal{O}(\rho^{-k})$ if $\rho^k f$ admits a smooth extension. We will sometimes paraphrase this as “ $\rho^k f$ is smooth up to the boundary”.

If $\rho : U \rightarrow \mathbb{R}$ is a defining function then any other defining function on U can be written as $\hat{\rho} = e^\alpha \rho$ for some smooth function $\alpha : U \rightarrow \mathbb{R}$. This readily implies that all the concepts introduced so far are independent of the choice of defining function. Therefore, we will often not specify the domain of definition of a defining function and say that $f = \mathcal{O}(\rho^\ell)$ for $\ell \neq 0$ without specifying ρ . We will also extend this terminology to other geometric objects like tensor fields and differential forms.

If $f = \mathcal{O}(\rho^{-k})$, then $\rho^k f$ has a well defined boundary value, which is a smooth function $U \cap \partial M \rightarrow \mathbb{R}$. This boundary value *does* depend on the choice of defining function, moving from ρ to $\hat{\rho} = e^\alpha \rho$ it gets rescaled by $e^{k\alpha}$. So here one has to be careful about the dependence on the defining function.

Definition 2.1. The sheaf of (local) vector fields on M will be denoted by $\mathfrak{X}$. This has a canonical sub-sheaf, that we denote by $\mathfrak{X}_\partial$, with $\mathfrak{X}_\partial(U)$ consisting of those $\xi \in \mathfrak{X}(U)$ which are tangent to ∂M along $U \cap \partial M$. So explicitly, we must have $\xi(x) \in T_x \partial M \subset T_x M$ for any $x \in U \cap \partial M$. For a defining function $\rho : U \rightarrow \mathbb{R}$, this may be equivalently characterized as $d\rho(\xi) = \mathcal{O}(\rho)$. Observe that $\mathfrak{X}_\partial(U)$ is a Lie subalgebra of $\mathfrak{X}(U)$. Indeed for $\xi, \eta \in \mathfrak{X}_\partial(U)$, the fact that $0 = dd\rho(\xi, \eta)$ immediately implies that $d\rho([\xi, \eta]) = \xi \cdot d\rho(\eta) - \eta \cdot d\rho(\xi)$. By assumption $d\rho(\eta) = \rho f$ for a smooth function $f : U \rightarrow \mathbb{R}$, so $\xi \cdot d\rho(\eta) = d\rho(\xi)f + \rho df(\xi) = \mathcal{O}(\rho)$ and likewise for the second summand.

2.2. The definition of projective pre-compactness. Let N be a smooth manifold and let ∇ and $\hat{\nabla}$ be linear connections on the tangent bundle TN . Recall that ∇ and $\hat{\nabla}$ are said to be *projectively equivalent* if they have the same geodesics up to parametrization. Since one can remove the torsion of a linear connection on TN without changing the geodesics, one usually works in the setting of torsion free connections. Then, the condition can be equivalently phrased as existence of a one-form $\Upsilon \in \Omega^1(N)$ such that

$$(2.1) \quad \hat{\nabla}_\xi \eta = \nabla_\xi \eta + \Upsilon(\xi)\eta + \Upsilon(\eta)\xi.$$

We will sometimes write this formally as $\hat{\nabla} = \nabla + \Upsilon$.

The concept of projective compactness was introduced in [9] as an analog of conformal compactness of (pseudo-)Riemannian metrics. In the setting $\overline{M} = M \cup \partial M$

from Section 2.1, conformal compactness is a condition on metrics g on M requiring that for one or equivalently any local defining function ρ for ∂M the metric $\rho^2 g$ admits a smooth extension to the boundary with non-degenerate boundary values.

Projective compactness of a linear connection ∇ on TM is defined in a similar spirit by requiring that for certain one-forms Υ (that are not smooth up to the boundary) the projective modifications $\nabla + \Upsilon$ admit a smooth extension to the boundary. The appropriate one-forms turn to be $\frac{d\rho}{\beta\rho}$ for local defining functions ρ for ∂M and a constant $\beta \in (0, 2]$ which is called the *order* of projective compactness. (Observe that β cannot be absorbed into a constant rescaling of ρ since we are dealing with a logarithmic derivative here.) In general, the cases $\beta = 1$ and $\beta = 2$ are the most interesting ones. While the basic idea of our extension could also be applied for other values of β we restrict to the case $\beta = 1$ in this article.

As in [9], we will mainly study the case of *special* linear connections here, i.e. connections that preserve a volume density. It is straightforward to see that any projective class admits such connections, and in the following when we mention a connection ∇ it may be assumed that it is special, unless we state otherwise. So for a local defining function $\rho : U \rightarrow \mathbb{R}$ for ∂M , we consider the one-form $\frac{d\rho}{\rho} \in \Omega^1(U \cap M)$ and for a special linear connection ∇ on TM , we consider the projectively modified linear connection $\nabla^\rho := \nabla + \frac{d\rho}{\rho}$ on $U \cap M$. Observe that for another defining function $\hat{\rho} = e^\alpha \rho$ we get $d\hat{\rho} = \hat{\rho}d\alpha + e^\alpha d\rho$ and hence $\frac{d\hat{\rho}}{\hat{\rho}} = \frac{d\rho}{\rho} + d\alpha$. Hence the connections ∇^ρ and $\nabla^{\hat{\rho}}$ are projectively related by the one-form $d\alpha$ that is smooth up to the boundary. This shows that if the conditions imposed in the following definition hold for one defining function on U then they hold for any defining function. So, loosely speaking, they are independent of the choice of defining function.

Definition 2.2. In our usual setting $\overline{M} = M \cup \partial M$, consider a special linear connection ∇ on TM and the associated modified connection $\nabla^\rho = \nabla + \frac{d\rho}{\rho}$ determined by any local defining function $\rho : U \rightarrow \mathbb{R}$.

(1) We say that ∇ is *projectively pre-compact* (of order 1) if and only if the following conditions are satisfied for each U and ρ :

- (i) $d\rho(\nabla_\xi^\rho \eta)$ is smooth up to the boundary for all $\xi, \eta \in \mathfrak{X}(U)$.
- (ii) $\rho \nabla_\xi^\rho \eta$ is smooth up to the boundary for all $\xi, \eta \in \mathfrak{X}(U)$.
- (iii) $\nabla_\xi^\rho \eta$ is smooth up to the boundary if at least one of the vector fields lies in $\mathfrak{X}_\partial(U) \subset \mathfrak{X}(U)$.

(2) We say that ∇ is *projectively compact* (of order 1) if $\nabla_\xi^\rho \eta$ is smooth up to the boundary for all $\xi, \eta \in \mathfrak{X}(U)$.

A pseudo-Riemannian metric g on M is called *projectively pre-compact*, or, respectively, *projectively compact*, (of order 1) if and only if its Levi-Civita connection is.

Thus projective pre-compactness should be interpreted as the fact that ∇^ρ “almost” admits a smooth extension to the boundary. Observe in particular, that we

can restrict any linear connection to operations $\mathfrak{X} \times \mathfrak{X}_\partial \rightarrow \mathfrak{X}$ and $\mathfrak{X}_\partial \times \mathfrak{X} \rightarrow \mathfrak{X}$ and projective modification makes sense in this setting. Condition (iii) then exactly says that the projective modifications of these “restricted connections” are smooth up to the boundary.

2.3. The canonical defining density. A major tool for the study of projectively compact linear connections is projective tractor calculus. In the setting of projective compactness, this is automatically available, since for a projectively compact linear connection ∇ , the projective structure on M defined by ∇ by definition admits a smooth extension to all of $\overline{M}$. Hence all tools from projective differential geometry are automatically available on all of $\overline{M}$ and several results on smooth extendability follows straight from projective invariance.

Things are not as straightforward in the projectively pre-compact case since one cannot expect the projective structure to extend to $\overline{M}$. For example, Theorem 3.3 of [9] analyzes the case of a Ricci-flat linear connection ∇ on TM which preserves some volume form on M . It is shown there that if the projective structure determined by ∇ extends smoothly to $\overline{M}$, but ∇ itself does not extend smoothly to a neighborhood of any boundary point, then ∇ has to be projectively compact of order 1.

The advantage we have here (compared with say conformal geometry) is that the bundles involved in projective tractor calculus do not need a projective structure to be defined. This is most easily seen by starting from the standard cotractor bundle $\mathcal{T}^*$, which can be defined as the first jet prolongation of a certain density bundle $\mathcal{E}(1)$ (that exists on any smooth manifold), see [4] and section 2.2 of [9] for the definitions of $\mathcal{E}(w)$ for $w \in \mathbb{R}$.

If ∇ is a special linear connection, then any of the density bundles $\mathcal{E}(w)$ admits a non-zero section that is parallel for (the connection induced by) ∇ and hence nowhere vanishing on M and this section is unique up to a constant multiple. The first step in [9] is then to prove that if ∇ is projectively compact of order 1, then any non-zero density $\sigma \in \mathcal{E}(1)$ which is parallel for ∇ admits a smooth extension by 0 to a defining density for ∂M . That means that for any connection $\hat{\nabla}$ in the projective class determined by ∇ and which is smooth up to the boundary, $\hat{\nabla}\sigma$ is nowhere vanishing on ∂M . The jet $j^1\sigma$ then defines a global smooth section $L(\sigma)$ of $\mathcal{T}^* = J^1\mathcal{E}(1)$, which is called the *scale tractor* associated to ∇ .

Now we can prove that these facts extend to the case where ∇ is projectively pre-compact, although the interpretation of $J^1\mathcal{E}(1)$ as a projective cotractor bundle is lost.

Proposition 2.3. *In our usual setting $\overline{M} = M \cup \partial M$ let ∇ be a special projectively pre-compact connection on TM and let $\sigma \in \Gamma(\mathcal{E}(1))$ be a nonzero density that is parallel for ∇ . Then σ smoothly extends by 0 to a defining density for ∂M . In particular, it determines a global smooth nowhere vanishing section $I := j^1\sigma \in \Gamma(J^1\mathcal{E}(1))$ on $\overline{M}$.*

Proof. Our strategy is to construct, locally near ∂M , a modification $\tilde{\nabla}$ of ∇ , which is projectively compact of order 1 and such that $\tilde{\nabla}\sigma = 0$. Then the claim follows

from applying Proposition 2.3 of [9] to $\tilde{\nabla}$. So we fix an open subset U that meets the boundary and a defining function $\rho : U \rightarrow \mathbb{R}$ for ∂M and consider the projective modification $\nabla^\rho = \nabla + \frac{d\rho}{\rho}$ of ∇ . Choosing arbitrary vector fields $\xi, \eta \in \mathfrak{X}(U)$, we know by definition that $\rho \nabla_\xi^\rho \eta$ admits a smooth extension of the boundary. Moreover, since $d\rho(\nabla_\xi^\rho \eta)$ itself is smooth up to the boundary, the boundary value of $\rho \nabla_\xi^\rho \eta$ is a smooth section of $T\partial M \subset T\overline{M}|_{\partial M}$. Now the expression $\rho \nabla_\xi^\rho \eta$ evidently is linear over smooth functions in ξ and one immediately verifies that it is linear over smooth functions in η up to $\mathcal{O}(\rho)$. Moreover, by definition $\rho \nabla_\xi^\rho \eta$ itself is $\mathcal{O}(\rho)$ if either ξ or η lies in $\mathfrak{X}_\partial(U) \subset \mathfrak{X}(U)$ and hence in particular if one of them is $\mathcal{O}(\rho)$.

Together, these facts imply that $(\xi, \eta) \mapsto \rho \nabla_\xi^\rho \eta|_{\partial M}$ is induced by applying a bundle map $T\overline{M}|_{\partial M} \times T\overline{M}|_{\partial M} \rightarrow T\partial M$ to the boundary values of ξ and η and we know in addition that this bundle map vanishes if one of its entries lies in $T\partial M \subset T\overline{M}|_{\partial M}$. Since the quotient bundle $T\overline{M}|_{\partial M}/T\partial M$ is trivialized by $d\rho$, we conclude that there is a smooth section ψ of $T\partial M$, such that the boundary value of $\rho \nabla_\xi^\rho \eta$ equals $d\rho(\xi)|_{\partial M} d\rho(\eta)|_{\partial M} \psi$.

Now we can extend ψ to a vector field $\tilde{\psi} \in \mathfrak{X}(U)$ such that $d\rho(\tilde{\psi}) \equiv 0$. Then $A := d\rho \otimes d\rho \otimes \tilde{\psi}$ is a smooth section of $(S^2 T^* \overline{M} \otimes T\overline{M})_0$, where the subscript indicates the trace-free part. Of course $\tilde{\nabla}_\xi \eta = \nabla_\xi \eta - \frac{1}{\rho} A(\xi, \eta)$ is a well-defined linear connection on $U \cap M$, which is torsion-free since A is symmetric. Moreover, since A is trace-free, σ is parallel for the connection induced by $\tilde{\nabla}$. Passing to the projective modifications of the two connections determined by ρ , we of course get $\tilde{\nabla}_\xi^\rho \eta = \nabla_\xi^\rho \eta - \frac{1}{\rho} A(\xi, \eta)$. By construction $\rho \tilde{\nabla}_\xi^\rho \eta$ admits a smooth extension to the boundary for any $\xi, \eta \in \mathfrak{X}(\overline{M})$ and again by construction this has vanishing boundary value. Thus $\tilde{\nabla}_\xi^\rho \eta$ admits a smooth extension to the boundary, which by definition implies that $\tilde{\nabla}$ is projectively compact of order one. $\square$

Having the canonical defining density σ at hand, we can actually convert the vector field ψ obtained in the proof into a canonical weighted object, which expresses the obstruction to ∇ being projectively compact. Knowing that σ is a defining density for ∂M , we conclude that for a linear connection $\hat{\nabla}$ on $\mathcal{E}(1)$, the derivative $\hat{\nabla}\sigma$ is independent of the choice of $\hat{\nabla}$ along the zero locus ∂M of σ . Since σ is a defining density, we also conclude that for any defining function ρ for the boundary, $\frac{\sigma}{\rho}$ admits a smooth extension to the boundary with nowhere vanishing boundary value. This readily implies that $\sigma \nabla_\xi^\rho \eta$ admits a smooth extension to the boundary for any $\xi, \eta \in \mathfrak{X}(\overline{M})$. Then the considerations in the proof above can be rephrased as the fact that there exists a section μ of $T\partial M(-1) := T\partial M \otimes \mathcal{E}(-1)$ such that the boundary value of $\sigma \nabla_\xi^\rho \eta$ equals $(\hat{\nabla}_\xi \sigma)|_{\partial M} (\hat{\nabla}_\eta \sigma)|_{\partial M} \mu$ for any ξ, η . The only dependence on a choice here is via the use of ∇^ρ . But as observed above for any other defining function $\hat{\rho}$, the difference $\nabla_\xi^\rho \eta - \nabla_\xi^{\hat{\rho}} \eta$ admits a smooth extension to the boundary, so $\mu \in \Gamma(T\partial M(-1))$ is canonically associated to ∇ .

2.4. Boundary structures. Let ∇ be a special linear connection on M and let $\sigma \in \Gamma(\mathcal{E}(1)|_M)$ be a nonzero density which is parallel for ∇ . In the case that

∇ is projectively compact of order 1, it was shown in Proposition 3.1 of [9] that σP_{ab} is smooth up to the boundary and defines a second fundamental form along the boundary. Here P_{ab} is the projective Schouten tensor which for a special connection is just a constant multiple of the Ricci curvature. In particular, this implies that the boundary is totally geodesic provided that ∇ has the property that its Ricci curvature admits a smooth extension to the boundary. In this case, one then obtains an induced projective structure on the boundary, which, in the case of Ricci-flat projectively compact metrics, is studied in detail in [9].

Thus we see that additional conditions will be necessary to obtain a boundary geometry induced by a connection ∇ on TM which is only projectively pre-compact. Surprisingly, not much is lost compared to the projectively compact case. We first discuss the connections on densities induced by the connections ∇^ρ . Recall that on a smooth manifold N of dimension $n + 1$, one can define the bundle $\text{vol}(N)$ of volume densities as used in Section 2.2 of [9] as the square root of the trivial line bundle $(\Lambda^{n+1}T^*M) \otimes (\Lambda^{n+1}T^*M)$. The density bundle $\mathcal{E}(1)$ which is relevant for our purposes is then defined as the $(n + 2)^{\text{nd}}$ root of $\text{vol}(N)$. In particular, in our setting of $\overline{M} = M \cup \partial M$, any linear connection on TM induces a linear connection on $\mathcal{E}(1)|_M$, and it is natural to ask whether such a connection is smooth up to the boundary.

Lemma 2.4. *In our usual setting of $\overline{M} = M \cup \partial M$, let ∇ be a special linear connection on TM , which is projectively pre-compact. Let $\rho : U \rightarrow \mathbb{R}$ be a local defining function for ∂M and ∇^ρ the corresponding modified connection. Then the induced connection ∇^ρ on $\mathcal{E}(1)|_{U \cap M}$ smoothly extends to all of U .*

Proof. Fixing a non-zero section $\sigma \in \Gamma(\mathcal{E}(1)|_U)$ which is parallel for ∇ , we know from Proposition 2.3 that σ smoothly extends by 0 to a defining density for $U \cap \partial M$. By definition, this means that there is a smooth, nowhere vanishing section $\sigma^\rho \in \Gamma(\mathcal{E}(1)|_U)$ such that σ coincides with $\rho\sigma^\rho$ on $U \cap M$. But over $U \cap M$, ∇^ρ is obtained from ∇ via a smooth projective change, so we can use the usual formula for the change of the induced connection on densities there. Hence for any vector field $\xi \in \mathfrak{X}(U \cap M)$, we get

$$0 = \nabla_\xi(\rho\sigma^\rho) = \nabla_\xi^\rho(\rho\sigma^\rho) - \frac{d\rho(\xi)}{\rho}\rho\sigma^\rho = \rho\nabla_\xi^\rho\sigma^\rho.$$

Since ρ is nowhere vanishing on $U \cap M$, we conclude that $\nabla_\xi^\rho\sigma^\rho = 0$ on $U \cap M$.

But now any section τ of $\mathcal{E}(1)|_U$ can be written as $f\sigma^\rho$ for a function $f \in C^\infty(U, \mathbb{R})$ and thus for $\xi \in \mathfrak{X}(U)$, we get $\nabla_\xi^\rho\tau = df(\xi)\sigma^\rho$ which is smooth up to the boundary. $\square$

This easily brings us to a canonical weighted second fundamental form and indeed a bit further than that. Starting from ∇ , we can take the smooth section $\sigma \in \Gamma(\mathcal{E}(1))$ and then form $\nabla^\rho\sigma$ which is a well defined smooth section of $T^*M(1)$.

Theorem 2.5. *The section $\nabla^\rho\nabla^\rho\sigma$ of $\otimes^2T^*M(1)$ is symmetric and admits a smooth extension to the boundary. Denoting the boundary value by Φ^ρ , $\Phi^\rho(\xi, \eta)$ is independent of ρ if at least one of the vector fields ξ, η lies in $\Gamma(T\partial M) \subset$*

$\Gamma(TM|_{\partial M})$, so we denote it by Φ in this case. If $\xi \in \mathfrak{X}(\overline{M})$ and $\eta \in \mathfrak{X}_\partial(\overline{M})$, then $\Phi(\xi|_{\partial M}, \eta|_{\partial M})$ is the boundary value of $\frac{d\rho(\xi)d\rho(\eta)}{\rho} - d\rho(\nabla_\xi^\rho \eta)$. If also $\xi \in \mathfrak{X}_\partial(M)$, then the first summand vanishes along the boundary, so $\Phi|_{T\partial M \times T\partial M}$ defines a canonical weighted second fundamental form for ∂M .

Proof. Take vector fields $\xi, \eta \in \mathfrak{X}(\overline{M})$. Over the interior, we get $\nabla^\rho \nabla^\rho \sigma(\xi, \eta) = \nabla_\xi^\rho \nabla_\eta^\rho \sigma - \nabla_{\nabla_\xi^\rho \eta}^\rho \sigma$. By Lemma 2.4, the first summand is smooth up to the boundary. In the proof of this Lemma, we have seen that $\nabla^\rho \sigma = d\rho \otimes \sigma^\rho$, so the second summand can be written as $d\rho(\nabla_\xi^\rho \eta) \sigma^\rho$, which by definition is smooth up to the boundary, too. This observation also shows that

$$(2.2) \quad \nabla^\rho \nabla^\rho \sigma(\xi, \eta) = (\xi \cdot d\rho(\eta) - d\rho(\nabla_\xi^\rho \eta)) \sigma^\rho.$$

In the interior, this expression is visibly symmetric since $dd\rho = 0$ and since ∇^ρ is torsion-free as a projective modification of the torsion-free connection ∇ .

For a second defining function $\hat{\rho} = e^f \rho$, we get $d\hat{\rho} = e^f d\rho + \hat{\rho} df$ and inserting this, we see that

$$(2.3) \quad \xi \cdot d\hat{\rho}(\eta) = e^f df(\xi) d\rho(\eta) + e^f \xi \cdot d\rho(\eta) + e^f d\rho(\xi) df(\eta) + \mathcal{O}(\rho)$$

On the other hand, $\nabla_\xi^{\hat{\rho}} \eta = \nabla_\xi^\rho \eta + df(\xi) \eta + df(\eta) \xi$, and hence $-d\hat{\rho}(\nabla_\xi^{\hat{\rho}} \eta)$ can be written as

$$(2.4) \quad -e^f (d\rho(\nabla_\xi^\rho \eta + df(\xi) \eta + df(\eta) \xi) + df(\rho \nabla_\xi^\rho \eta)) + \mathcal{O}(\rho).$$

Finally, since σ^ρ is characterized by $\rho \sigma^\rho = \sigma$, we see that $\sigma^{\hat{\rho}} = e^{-f} \sigma^\rho$, so using (2.2), we see that

$$\nabla^{\hat{\rho}} \nabla^{\hat{\rho}} \sigma(\xi, \eta) = \nabla^\rho \nabla^\rho \sigma(\xi, \eta) - df(\rho \nabla_\xi^\rho \eta) \sigma^\rho + \mathcal{O}(\rho).$$

If either ξ or η lies in $\mathfrak{X}_\partial(\overline{M})$, then $\rho \nabla_\xi^\rho \eta = \mathcal{O}(\rho)$, hence in this case ρ and $\hat{\rho}$ lead to the same boundary value. Finally, if η lies in $\mathfrak{X}_\partial(\overline{M})$, then $\xi \cdot d\rho(\eta) = \frac{d\rho(\xi)d\rho(\eta)}{\rho} + \mathcal{O}(\rho)$ and if also ξ lies in $\mathfrak{X}_\partial(\overline{M})$, this itself is $\mathcal{O}(\rho)$, which completes the proof. $\square$

2.5. The case of totally geodesic boundary. We next study vanishing of (parts of) the canonical bilinear form Φ from Theorem 2.5.

Proposition 2.6. *For $\overline{M} = M \cup \partial M$ let ∇ be a special linear connection on TM which is projectively pre-compact and has the property that the weighted second fundamental form $\Phi|_{T\partial M \times T\partial M}$ from Theorem 2.5 vanishes identically.*

Then for any local defining function ρ , the connection ∇^ρ restricts to give a torsion-free linear connection on $T\partial M \rightarrow \partial M$ and any other defining function $\hat{\rho}$ leads to a projectively equivalent connection. Hence the boundary ∂M inherits a projective structure from ∇ .

Proof. By Theorem 2.5 vanishing of Φ on $T\partial M \times T\partial M$ is equivalent to the fact that for $\xi, \eta \in \mathfrak{X}_\partial(\overline{M})$, $d\rho(\nabla_\xi^\rho \eta)$ vanishes along the boundary, so ∇^ρ maps $\mathfrak{X}_\partial(\overline{M}) \times \mathfrak{X}_\partial(\overline{M})$ to $\mathfrak{X}_\partial(\overline{M})$. Given vector fields $\underline{\xi}$ and $\underline{\eta}$ on ∂M , we can choose smooth extensions to $\xi, \eta \in \mathfrak{X}_\partial(\overline{M})$ and then consider the boundary value of $\nabla_\xi^\rho \eta$ which

lies in $\mathfrak{X}(\partial M)$. One immediately verifies that this depends only on ξ and η and not on the choice of the extensions. Thus we obtain a well defined operation $\mathfrak{X}(\partial M) \times \mathfrak{X}(\partial M) \rightarrow \mathfrak{X}(\partial M)$, which clearly defines a linear connection on $T\partial M$.

Torsion-freeness of ∇^ρ then immediately implies torsion-freeness of the induced connection. Finally, we have seen already that for $\hat{\rho} = e^f \rho$, we get $\nabla_\xi^{\hat{\rho}} \eta = \nabla_\xi^\rho \eta + df(\xi)\eta + df(\eta)\xi$. Since the function f is smooth up to the boundary, so is df and the restriction of df to $T\partial M$ defines an element of $\Omega^1(\partial M)$, which describes a projective equivalence between the induced connections on the boundary. $\square$

Of course, there is a natural strengthening of this condition, which will turn out to be of crucial importance in what follows, namely that Φ vanishes identically on all of $T\overline{M}|_{\partial M} \times T\partial M$. By Theorem 2.5, this is the case if and only if for any $\xi \in \mathfrak{X}(\overline{M})$ and $\eta \in \mathfrak{X}_\partial(\overline{M})$, $d\rho(\nabla_\xi^\rho \eta)$ and $\frac{d\rho(\xi)d\rho(\eta)}{\rho}$ have the same boundary value.

Definition 2.7. (1) If ∇ is projectively pre-compact and is such that the weighted second fundamental form $\Phi|_{T\partial M \times T\partial M}$ from Theorem 2.5 vanishes, then we say that ∇ is *projectively pre-compact with totally geodesic boundary*.

(2) In the situation of (1) for any of the connections ∇^ρ , we will denote the induced linear connection on the boundary also by ∇^ρ .

(3) If ∇ is projectively pre-compact and such that the bilinear form Φ from Theorem 2.5 vanishes on $TM|_{\partial M} \times T\partial M$, then we say that ∇ is *projectively pre-compact with strongly totally geodesic boundary*.

2.6. Ricci curvature and second fundamental form. We next analyze the curvature asymptotics of the modified connections ∇^ρ . Recall that for a smooth manifold N and a general linear connection on TN , there is a Ricci-type contraction of the curvature, which defines a section of $\otimes^2 T^*N$. This is symmetric if and only if the connection preserves a volume density.

Lemma 2.8. For $\overline{M} = M \cup \partial M$ consider a special linear connection ∇ on TM which is projectively pre-compact. For a local defining function ρ for ∂M , consider the modified connection ∇^ρ on TM and let R^ρ be its curvature and Ric^ρ its Ricci curvature.

(1) For $\xi, \eta, \zeta \in \mathfrak{X}(\overline{M})$ the vector field $\rho R^\rho(\xi, \eta)(\zeta)$ admits a smooth extension to the boundary. Moreover, $R^\rho(\xi, \eta)(\zeta)$ itself admits a smooth extension to the boundary if both ξ and η are in $\mathfrak{X}_\partial(\overline{M})$.

(2) For $\xi, \eta \in \mathfrak{X}(\overline{M})$, the function $\rho \text{Ric}^\rho(\xi, \eta)$ admits a smooth extension to the boundary. Moreover, $\text{Ric}^\rho(\xi, \eta)$ admits a smooth extension to the boundary if both ξ and η lie in $\mathfrak{X}_\partial(\overline{M})$.

(3) Suppose that ∇ is projectively pre-compact with totally geodesic boundary. Then $R^\rho(\xi, \eta)(\zeta)$ admits a smooth extension to the boundary if ζ lies in $\mathfrak{X}_\partial(\overline{M})$ and $\text{Ric}^\rho(\xi, \eta)$ also admits a smooth extension to the boundary if one of its entries lies in $\mathfrak{X}_\partial(\overline{M})$.

Proof. We directly use the defining equation

$$(2.5) \quad R^\rho(\xi, \eta)(\zeta) = \nabla_\xi^\rho \nabla_\eta^\rho \zeta - \nabla_\eta^\rho \nabla_\xi^\rho \zeta - \nabla_{[\xi, \eta]}^\rho \zeta.$$

Observe that by skew symmetry of R^ρ in ξ and η , we may always assume that one of the two fields, say η , satisfies $d\rho(\eta) \equiv 0$. (Locally fix a vector field ξ_0 such that $d\rho(\xi_0) \equiv 1$ and then expand $\xi = d\rho(\xi)\xi_0 + (\xi - d\rho(\xi)\xi_0)$ and like-wise for η .)

In the interior, we have $\rho R^\rho(\xi, \eta)(\zeta) = R^\rho(\xi, \eta)(\rho\zeta)$ and since $\rho\zeta$ lies in $\mathfrak{X}_\partial(\overline{M})$, replacing ζ by $\rho\zeta$ in (2.5), the last summand in the right hand side admits a smooth extension to the boundary. Likewise, $\nabla_\xi^\rho \rho\zeta$ admits a smooth extension and since $\eta \in \mathfrak{X}_\partial(\overline{M})$, we see that $\nabla_\eta^\rho \nabla_\xi^\rho \rho\zeta$ admits a smooth extension to the boundary. Finally, by our assumption on η , we have $\nabla_\eta^\rho \rho\zeta = \rho \nabla_\eta^\rho \zeta$ and since this is in $\mathfrak{X}_\partial(\overline{M})$, $\nabla_\xi^\rho \nabla_\eta^\rho \rho\zeta$ also is smooth up to the boundary. Hence $\rho R^\rho(\xi, \eta)(\zeta)$ always admits a smooth extension to the boundary.

If both ξ and η lie in $\mathfrak{X}_\partial(\overline{M})$, then so does $[\xi, \eta]$, and hence $\nabla_\xi^\rho \zeta$, $\nabla_\eta^\rho \zeta$ and $\nabla_{[\xi, \eta]}^\rho \zeta$ all admit smooth extensions. This in turn shows that also $\nabla_\xi^\rho \nabla_\eta^\rho \zeta$ and $\nabla_\eta^\rho \nabla_\xi^\rho \zeta$ extend smoothly, and this completes the proof of (1).

In the setting of (3), assume that ζ and without loss of generality η lie in $\mathfrak{X}_\partial(\overline{M})$. Then the last two summands in (2.5) cause no problems while the assumptions imply that $\nabla_\eta^\rho \zeta$ lies in $\mathfrak{X}_\partial(\overline{M})$ so $\nabla_\xi^\rho \nabla_\eta^\rho \zeta$ admits a smooth extension to the boundary, too.

Switching to the Ricci curvature, the fact that ρRic^ρ admits a smooth extension to the boundary follows immediately from (1) since Ric^ρ is a contraction of R^ρ . For the second part, this is not as easy, since when computing $\text{Ric}^\rho(\xi, \eta)$ as a trace, ξ and η are not both inserted into the form slots of R^ρ . As in (1), we locally choose $\xi_0 \in \mathfrak{X}(\overline{M})$ such that $d\rho(\xi_0) \equiv 1$ and extend this to a local frame by $\xi_i \in \mathfrak{X}(\overline{M})$, $i = 1, \dots, n$ such that $d\rho(\xi_i) \equiv 0$ for $i \geq 1$. Then the dual coframe has the form $d\rho, \alpha_i$ for $i = 1, \dots, n$, and by definition we get (on M)

$$(2.6) \quad \text{Ric}^\rho(\xi, \eta) = d\rho(R^\rho(\xi, \xi_0)(\eta)) + \sum_{i \geq 1} \alpha_i(R^\rho(\xi, \xi_i)(\eta)).$$

Since $\xi, \xi_i \in \mathfrak{X}_\partial(\overline{M})$ each term in the sum in the right hand side admits a smooth extension to the boundary. For the first term in the right hand side, we have to go back to the defining equation (2.5). But by construction $\nabla_{[\xi, \xi_0]}\eta$, $\nabla_{\xi_0}\eta$ and hence $\nabla_\xi \nabla_{\xi_0}\eta$, and $\nabla_\eta \nabla_\xi \eta$ admit smooth extension to the boundary. But this implies that also $d\rho(\nabla_{\xi_0} \nabla_\xi \eta)$ admits a smooth extension, so the result follows.

Under the stronger assumption of (3), we know from above that all curvature terms in the right hand side of (2.6) admit a smooth extension to the boundary if η lies in $\mathfrak{X}_\partial(\overline{M})$. By symmetry of Ric^ρ in the interior, this implies the last part of (3). $\square$

Recall that the Schouten tensor of a linear connection on the tangent bundle TN of a smooth manifold N is an equivalent encoding of the Ricci curvature, which mixes the symmetric and skew symmetric parts with different factors. In particular, for a special connection, it is just $\frac{1}{\dim(N)-1}$ times the Ricci curvature. Thus we see that the analog of the statements on Ric^ρ from Lemma 2.8 also hold for the Schouten tensor P^ρ of ∇^ρ .

Theorem 2.9. *Consider $\overline{M} = M \cup \partial M$ and a special connection ∇ on TM which is projectively pre-compact. Let $\sigma \in \Gamma(\mathcal{E}(1)|_M)$ be a nonzero section that is parallel*

for ∇ and let P be the Schouten tensor of ∇ . Then the section σP of $S^2T^*M(1)$ admits a smooth extension to the boundary and its boundary value restricts on $T\partial M \times T\partial M$ to the canonical weighted second fundamental form Φ from Theorem 2.5. Moreover, the form Φ vanishes on $T\overline{M}|_{\partial M} \times T\partial M$ if and only if the boundary value of σP has the same property.

Proof. On M , we can write $\sigma P = \nabla \nabla \sigma + \sigma P$ and it is well known that the right hand side defines a projectively invariant second order differential operator $\Gamma(\mathcal{E}(1)) \rightarrow \Gamma(S^2T^*M(1))$, see [4]. Hence we may as well express it in terms of the projectively equivalent connection ∇^ρ , whence

$$(2.7) \quad \sigma P = \nabla^\rho \nabla^\rho \sigma + \sigma P^\rho.$$

Now the two terms on the right hand side admit a smooth extension to the boundary. For the first term this follows from Theorem 2.5 and for the second term this follows from Lemma 2.8, taking into account that $\sigma = \rho \sigma^\rho$. This proves the first claim. For the second claim, we know that for $\xi, \eta \in \mathfrak{X}_\partial(\overline{M})$, the function $P^\rho(\xi, \eta)$ admits a smooth extension to the boundary (again using Lemma 2.8), so $\sigma P^\rho(\xi, \eta)$ vanishes along the boundary. Thus $\sigma P(\xi, \eta) = \nabla^\rho \nabla^\rho \sigma(\xi, \eta)$ and the boundary value of this is the canonical weighted second fundamental form from Theorem 2.5. Hence if σP vanishes on $T\partial M \times T\partial M$, then we are in the case of totally geodesic boundary, so by part (3) of Lemma 2.8, $P^\rho(\xi, \eta)$ admits a smooth extension the boundary if one of its entries lies in $\mathfrak{X}_\partial(\overline{M})$. Hence we see that, in this situation, the boundary value of σP again coincides with Φ . $\square$

Corollary 2.10. *For $\overline{M} = M \cup \partial M$, let ∇ be a linear connection on M that is projectively pre-compact and let P be its Schouten tensor. Then the boundary is totally geodesic if and only if $P(\xi, \eta)$ admits a smooth extension to the boundary whenever $\xi, \eta \in \mathfrak{X}_\partial(\overline{M})$ and strongly totally geodesic if and only if this holds if one of the two fields lies in $\mathfrak{X}_\partial(\overline{M})$.*

Moreover, if ∇ is projectively pre-compact with strongly geodesic boundary, then for any $\xi, \eta \in \mathfrak{X}(\overline{M})$, we get

$$(2.8) \quad d\rho(\nabla_\xi^\rho \eta) = \xi \cdot d\rho(\eta) + \rho P^\rho(\xi, \eta) - \rho P(\xi, \eta).$$

Proof. The first part follows immediately from Theorem 2.9, while (2.8) follows readily from combining (2.2) with (2.7). $\square$

2.7. Some b -calculus. For the next step we use the description of the sheaf $\mathfrak{X}_\partial$ on $\overline{M}$ developed by the Melrose school in geometric analysis. Since $\mathfrak{X}_\partial$ is a locally free sheaf, there is a smooth vector bundle $T^b\overline{M} \rightarrow \overline{M}$, called the b -tangent bundle, whose sheaf of smooth sections coincides with $\mathfrak{X}_\partial$. So by definition, a section ξ of $T^b\overline{M}$ is a special vector field, but one has to carefully distinguish between the two points of view, in particular if boundary values are concerned. As a vector field, ξ has a boundary value which is a vector field on ∂M , i.e. a smooth section of $T\partial M$. But ξ also has a boundary value as a section of $T^b\overline{M}$, which is a section of $T^b\overline{M}|_{\partial M}$, a bundle whose rank is one larger than the one of $T\partial M$. Indeed, mapping $\xi \in \Gamma(T^b\overline{M})$ to its boundary value as a vector field defines a surjective bundle map $T^b\overline{M}|_{\partial M} \rightarrow T\partial M$. A neat way to express things is that for $\xi \in \mathfrak{X}(M)$,

we have $\xi \in \Gamma(T^b\overline{M})$ if and only if $d\rho(\xi)|_{\partial M} = 0$ for one or equivalently any local defining function ρ i.e. iff $d\rho(\xi) = \mathcal{O}(\rho)$. And the boundary value of ξ as a section of $T^b\overline{M}$ vanishes iff $\xi = \rho\xi_1$ for some $\xi_1 \in \Gamma(T^b\overline{M})$ and hence iff $d\rho(\xi) = \mathcal{O}(\rho^2)$.

Lemma 2.11. *There is a short exact sequence*

$$(2.9) \quad 0 \rightarrow \partial M \times \mathbb{R} \rightarrow T^b\overline{M}|_{\partial M} \rightarrow T\partial M \rightarrow 0.$$

A canonical trivializing section of the first bundle can be realized as the boundary value of $\rho\nu \in \Gamma(T^b\overline{M})$, where ρ is a local defining function for the boundary and $\nu \in \mathfrak{X}(\overline{M})$ is a vector field such that $d\rho(\nu)|_{\partial M} \equiv 1$.

Proof. By construction, the vector field $\rho\nu$ vanishes along ∂M , so $\rho\nu \in \Gamma(T^b\overline{M})$. Since $d\rho(\rho\nu) = \rho + \mathcal{O}(\rho^2)$ and this is not $\mathcal{O}(\rho^2)$, we conclude that the boundary value of $\rho\nu$ as a section of $T^b\overline{M}$ spans the kernel of $T^b\overline{M}|_{\partial M} \rightarrow T\partial M$, and it remains to show that this boundary value is independent of the choice of ρ and ν .

Any other choice for ν is of the form $\hat{\nu} = \nu + \eta$, where $\eta \in \mathfrak{X}(\overline{M})$ is such that $d\rho(\eta)|_{\partial M} \equiv 0$. But then $\eta \in \Gamma(T^b\overline{M})$ and hence $\rho\hat{\nu} = \rho\nu + \rho\eta$ has the same boundary value as $\rho\nu$. Any other defining function is of the form $\hat{\rho} = e^f\rho$ and then $d\hat{\rho}|_{\partial M} = e^f d\rho|_{\partial M}$. Hence given a choice ν for ρ , we can put $\hat{\nu} = e^{-f}\nu$ and then $\hat{\rho}\hat{\nu} = \rho\nu$, which completes the proof. $\square$

Once the vector bundle $T^b\overline{M}$ is at hand, we can apply all the standard tools from differential geometry to it. This immediately allows us to form the weighted b -tangent bundle of weight $w \in \mathbb{R}$ by putting $T^b\overline{M}(w) := T^b\overline{M} \otimes \mathcal{E}(w)$. We can also apply vector bundle constructions, e.g. form the dual bundle $(T^b\overline{M})^*$, the symmetric square $S^2 T^b\overline{M}$ or the symmetric square $S^2(T^b\overline{M})^*$ of the dual. For the bundles constructed from $T^b\overline{M}$ (without dualization) the interpretation of sections is rather easy starting from tensor products of sections of $T^b\overline{M}$, which locally span the space of all sections. For example, a section b of $S^2 T^b\overline{M}$ is a section of $S^2 TM$, which contracted with $d\rho$ gives a section of $T^b\overline{M}$ that is $\mathcal{O}(\rho)$ and hence contracting b with two copies of $d\rho$ the result is a function that is $\mathcal{O}(\rho^2)$.

Things are a bit more subtle for the dual. As we have seen above, locally around a boundary point, we can form a local frame for $T^b\overline{M}$ consisting of $\rho\nu$ and vector fields $\zeta_1, \dots, \zeta_n$ such that $d\rho(\zeta_i) \equiv 0$. Now we can choose $\alpha_i \in \Omega^1(M)$ such that $\alpha_i(\nu) = 0$ and $\alpha_i(\zeta_j) = \delta_{ij}$ and add to this the form $\frac{d\rho}{\rho}$ on M . Since for $\xi \in \Gamma(T^b\overline{M})$, $d\rho(\xi) = \mathcal{O}(\rho)$, the function $\frac{d\rho(\xi)}{\rho}$ admits a smooth extension to the boundary. Thus $\frac{d\rho}{\rho}$ defines a smooth section of $(T^b\overline{M})^*$ over all of $\overline{M}$ and clearly the α_i also define such sections, so together these form the dual of the frame $\{\rho\nu, \zeta_1, \dots, \zeta_n\}$. This leads to a neat description of sections of $(T^b\overline{M})^*$: A section of this bundle is a one-form φ on M such that $\rho\varphi$ admits a smooth extension to the boundary and such that $\varphi(\xi)$ admits a smooth extension to the boundary if $\xi \in \mathfrak{X}_\partial(\overline{M})$.

In particular, we have $\Omega^1(\overline{M}) \subset \Gamma((T^b\overline{M})^*)$ corresponding to the inclusion $T^*\overline{M} \hookrightarrow (T^b\overline{M})^*$ dual to the projection $T^b\overline{M} \rightarrow T\overline{M}$. The elements in this subspace can be easily characterized. Choose a (local) defining function ρ and a vector field ν such that $d\rho(\nu)|_{\partial M} \equiv 1$. Then $\rho\nu \in \Gamma(T^b\overline{M})$ and we claim that if $\varphi \in \Gamma((T^b\overline{M})^*)$ satisfies $\varphi(\rho\nu)|_{\partial M} = 0$ then $\varphi \in \Omega^1(\overline{M})$. Indeed, we know

that $\rho\varphi \in \Omega^1(\overline{M})$, so it suffices to prove that this vanishes along the boundary. But $\xi \in \mathfrak{X}(M)$ can be written as $\xi_1 + d\rho(\xi)\nu$ where $\xi_1 := \xi - d\rho(\xi)\nu$ satisfies $d\rho(\xi_1) = \mathcal{O}(\rho)$ and hence $\xi_1 \in \mathfrak{X}_\partial(\overline{M})$. Thus $\varphi(\xi_1)$ admits a smooth extension to the boundary so $\rho\varphi(\xi_1)$ vanishes along the boundary. On the other hand $\rho\varphi(d\rho(\xi)\nu) = d\rho(\xi)\varphi(\rho\nu)$, so this is $\mathcal{O}(\rho)$ by assumption.

Having these descriptions at hand, the passage to the results of tensorial constructions is easy starting from tensor products of sections. For example, a section b of $S^2(T^b\overline{M})^*$ is a section of S^2T^*M such that ρ^2b admits a smooth extension to the boundary and for $\xi \in \mathfrak{X}_\partial(\overline{M})$, $\rho b(\xi, \cdot)$ admits a smooth extension to the boundary as a section of $(T^b\overline{M})^*$. Hence in particular for $\eta \in \mathfrak{X}_\partial(\overline{M})$, $b(\xi, \eta)$ admits a smooth extension to the boundary. As above, we have $\Gamma(S^2T^*\overline{M}) \subset \Gamma(S^2(T^b\overline{M})^*)$ and one immediately shows that a section b lies in this subspace if for ρ and ν as above and $\xi \in \mathfrak{X}_\partial(\overline{M})$, we have $b(\rho\nu, \xi) = \mathcal{O}(\rho)$ and $b(\rho\nu, \rho\nu) = \mathcal{O}(\rho^2)$.

2.8. A canonical connection. We can consider linear connections on bundles constructed from $T^b\overline{M}$, to which all the general results about linear connections can be applied. In particular, starting from a projectively pre-compact linear connection ∇ on $M \subset \overline{M}$, we have an induced connection on any density bundle $\mathcal{E}(w)$ by Proposition 2.3. Hence over M , we have an induced linear connection on any weighted tangent bundle, which we also denote by ∇ , and we obtain the following result.

Theorem 2.12. *For $\overline{M} = M \cup \partial M$, let ∇ be a projectively pre-compact linear connection on TM . Then for $\xi \in \mathfrak{X}(\overline{M})$ and $\eta \in \Gamma(T^b\overline{M}(-1)) \subset \Gamma(T\overline{M}(-1))$, the weighted vector field $\nabla_\xi\eta$ defined on M admits a smooth extension to the boundary.*

If the boundary is strongly totally geodesic, then this extension lies in $\mathfrak{X}_\partial(\overline{M})$. Hence, under this assumption, ∇ extends smoothly to a linear connection on the vector bundle $T^b\overline{M}(-1) \rightarrow \overline{M}$.

Proof. The usual formula for projective changes in [4] shows that, over M , we can write $\nabla_\xi\eta = \nabla_\xi^\rho\eta - \frac{d\rho(\eta)}{\rho}\xi$ (and the weight -1 is essential for the term containing $d\rho(\xi)$ to drop out). Since η is tangent to the boundary along the boundary, both summands admit a smooth extension to the boundary. Applying $d\rho$ to the right hand side, we get $d\rho(\nabla_\xi^\rho\eta) - \frac{d\rho(\xi)d\rho(\eta)}{\rho}$ which, in the case of strongly totally geodesic boundary, vanishes along the boundary (see Theorem 2.5). $\square$

Hence we have obtained a vector bundle $T^b\overline{M}(-1) \rightarrow \overline{M}$ of rank $n+1$, which is endowed with a linear connection that we still denote by ∇ . This bundle can be restricted to ∂M without problems, and it follows from the general theory of linear connections that one can also restrict the connection to the boundary. Indeed for $\xi \in \mathfrak{X}(\partial M)$ and $\eta \in \Gamma(T^b\overline{M}(-1)|_{\partial M})$ one chooses (local) extensions to $\tilde{\xi} \in \mathfrak{X}_\partial(\overline{M})$ and $\tilde{\eta} \in \Gamma(T^b\overline{M}(-1))$ and then considers $\nabla_{\tilde{\xi}}\tilde{\eta}|_{\partial M}$. It is clear that this depends only on $\tilde{\xi}|_{\partial M} = \xi$, while for η , the freedom is to add a $\rho\tilde{\eta}_1$ for $\eta_1 \in \Gamma(T^b\overline{M}(-1))$, and one immediately concludes that this leads to the same restriction to the boundary.

2.9. Relation to boundary tractors. Starting from a projectively pre-compact connection ∇ with strongly totally geodesic boundary, we get an induced projective structure on the boundary ∂M as well as the rank $n + 1$ vector bundle $T^b\overline{M}(-1)|_{\partial M} \rightarrow \partial M$ which is endowed with a linear connection induced by ∇ . One may hope that this is related to the projective standard tractor bundle of the boundary projective structure. There is a relation without additional assumptions, but the main result is that under a slightly stronger condition, the restriction indeed coincides with the standard tractor bundle and its normal tractor connection. We want to stress here that this conceptual description of the boundary tractors is new even in the case of projectively compact connections, the results in [9] apply only in the case that ∇ is Ricci flat in a neighborhood of the boundary. In contrast, we only require asymptotic Ricci-flatness in directions tangent to the boundary here.

Let us briefly recall the standard approach to projective tractors on a manifold N of dimension n endowed with a projective class of affine connections, see [4] for more details and explicit formulae. One usually starts from the standard cotractor bundle, which can be simply defined as a first jet prolongation of a line bundle $\mathcal{T}^*N := J^1\mathcal{E}(1)$. Here $\mathcal{E}(1) \rightarrow N$ is the bundle of (projective) 1-densities, a trivial line bundle characterized by the fact that $\mathcal{E}(1)^{\otimes 2n+2} \cong (\Lambda^n TN)^{\otimes 2}$. The jet exact sequence for this bundle has the form

$$0 \rightarrow T^*N(1) \rightarrow \mathcal{T}^*N \rightarrow \mathcal{E}(1) \rightarrow 0.$$

Any linear connection on $\mathcal{E}(1)$ defines a splitting of this short exact sequence, and in particular, one can apply this to the connections induced by the affine connections in the projective class. Hence any connection in the projective class gives rise to an identification $\mathcal{T}^*N \cong \mathcal{E}(1) \oplus T^*N(1)$ and one can use the connection defining the identification, its Schouten tensor and algebraic ingredients to define a linear connection on $\mathcal{T}^*N$ in the given splitting and then verify that the construction is independent of the choice of connection. Hence it gives rise to a canonical connection on $\mathcal{T}^*N$ called the *standard cotractor connection*.

The standard tractor bundle $\mathcal{T}N \rightarrow N$ is then defined as the dual bundle to $\mathcal{T}^*N$, which implies that one gets an exact sequence

$$(2.10) \quad 0 \rightarrow \mathcal{E}(-1) \rightarrow \mathcal{T}N \rightarrow TN(-1) \rightarrow 0$$

and any connection in the projective class determines a splitting of this sequence. The standard tractor connection on $\mathcal{T}N$ is then obtained from dualizing the standard cotractor connection and it is easy to derive the formula for the standard tractor connection in such a splitting, see [4].

Theorem 2.13. *For $\overline{M} = M \cup \partial M$, let ∇ be an affine connection on M , which is projectively pre-compact and let P be the Schouten tensor of ∇ . Suppose further that for all $\xi, \eta \in \mathfrak{X}(\overline{M})$, the function $P(\xi, \eta)$ admits a smooth extension to the boundary if at least one of the fields lies in $\mathfrak{X}_\partial(\overline{M})$ and its boundary value vanishes if both fields lie in $\mathfrak{X}_\partial(\overline{M})$. Then the bundle $T^b\overline{M}(-1)|_{\partial M}$ with the connection induced by ∇ can be naturally identified with the projective standard tractor bundle of the induced projective structure on ∂M and its normal tractor connection.*

Proof. First note that our assumptions imply that ∇ has strongly totally geodesic boundary, so we indeed get a linear connection on $T^b\overline{M}(-1)|_{\partial M}$ induced by ∇ . We next observe that by expression (2.9) from Lemma 2.11, the bundle $T^b\overline{M}(-1)|_{\partial M}$ fits into an exact sequence which looks like the sequence (2.10) for the standard tractor bundle. The only difference is that the (-1) -density bundle of ∂M is replaced by the restriction $\mathcal{E}(-1)|_{\partial M}$ of the (-1) -density bundle of $\overline{M}$. (Note that the power in the definition of (-1) -densities as a root of the bundle of volume forms depends on the dimension). So we first prove that there is a natural way to identify these two bundles.

Fix a non-zero density $\sigma \in \Gamma(\mathcal{E}(1)|_M)$ which is parallel for ∇ . By Proposition 2.3, this extends by 0 to a smooth section of $\mathcal{E}(1) \rightarrow \overline{M}$ which is a defining density for ∂M . Taking any linear connection $\tilde{\nabla}$ on $\mathcal{E}(1)$, the derivative $\tilde{\nabla}\sigma|_{\partial M}$ is a nowhere vanishing section of $T^*\partial M \otimes \mathcal{E}(1)|_{\partial M}$. This is independent of the choice of $\tilde{\nabla}$ and its kernel in a point $x \in \partial M$ coincides with $T_x\partial M \subset T_x\overline{M}$. Putting $n := \dim(\partial M)$, taking the wedge product with this weighted one form induces an isomorphism $\Lambda^n T^*\partial M \rightarrow \Lambda^{n+1} T^*\overline{M} \otimes \mathcal{E}(1)$, and hence between the squares of this line bundles. But the square of the first bundle is the (intrinsic) density bundle $\mathcal{E}_{\partial M}(-2n-2)$ while square of the second bundle is the restriction to ∂M of $\mathcal{E}(-2n-4) \otimes \mathcal{E}(2) = \mathcal{E}(-2n-2)$. Since this extends to powers, roots and duals, we can in particular identify $\mathcal{E}(1)|_{\partial M}$ with the $\mathcal{E}_{\partial M}(1)$, so we drop the subscript.

Thus we have an exact sequence of the right form. In order to obtain the usual formula for the standard tractor connection, we have to make a small change, however. We define the inclusion $\mathcal{E}(-1) \hookrightarrow T^b\overline{M}(-1)|_{\partial M}$ by sending a density τ to the boundary value of $-\tau\rho\nu$ where $d\rho(\nu)|_{\partial M} \equiv 1$. From the proof of Lemma 2.11 we also see how a choice of local defining function ρ gives rise to a splitting of the exact sequence. Given $\tilde{\eta} \in \Gamma(T^b\overline{M}(-1))$ the (-1) -density $-\frac{d\rho(\tilde{\eta})}{\rho}$ admits a smooth extension to the boundary and its boundary value depends only on the boundary value of $\tilde{\eta}$, which is a section of $T^b\overline{M}(-1)|_{\partial M}$. Hence we have constructed a bundle map $T^b\overline{M}(-1)|_{\partial M} \rightarrow \mathcal{E}(-1)$ (depending on the choice of ρ), which evidently splits the inclusion defined above.

So it remains to compute the connection induced by ∇ in such a splitting and we fix the defining function ρ to determine the splitting. Then we take $\tilde{\eta} \in \Gamma(T^b\overline{M}(-1))$ and $\tilde{\xi} \in \mathfrak{X}(\overline{M})$ and denote by $\xi \in \mathfrak{X}(\partial M)$ the boundary value of the vector field $\tilde{\xi}$. Then we know that, in the splitting determined by ρ , the boundary values of $\tilde{\eta}$ corresponds to (η, τ) , where η is the boundary value of the vector field $\tilde{\eta}$ and τ is the boundary value of $-\frac{d\rho(\tilde{\eta})}{\rho}$. Now we compute

$$(2.11) \quad \nabla_{\tilde{\xi}}\tilde{\eta} = \nabla_{\xi}^{\rho}\tilde{\eta} - \frac{d\rho(\tilde{\eta})}{\rho}\tilde{\xi},$$

and we have to determine the boundary value of this section of $T^b\overline{M}(-1)$ in the splitting determined by ρ . In the first component, this simply is the boundary value as a vector field, which is just $\nabla_{\xi}^{\rho}\eta + \tau\xi$, where ∇^{ρ} also denotes the connection on $T\partial M(-1)$ induced by the connection ∇^{ρ} in the projective class. This agrees with the formula for the standard tractor connection in Section 3.2 of [4].

To deal with the other component, we need one more observation. For $\tilde{\xi}$ and another vector field $\tilde{\zeta} \in \mathfrak{X}_\partial(\overline{M})$, we can apply the formula (2.8), taking into account that $P(\tilde{\xi}, \tilde{\eta}) = \mathcal{O}(\rho)$ by assumption, to conclude that

$$d\rho(\nabla_{\tilde{\xi}}^\rho \tilde{\zeta}) = \tilde{\xi} \cdot d\rho(\tilde{\zeta}) + \rho P^\rho(\tilde{\xi}, \tilde{\zeta}) + \mathcal{O}(\rho^2).$$

This equation can be twisted by a density without problems, one only has to replace the derivative in the first term of the right hand side by $\nabla_{\tilde{\xi}}^\rho$, so

$$d\rho(\nabla_{\tilde{\xi}}^\rho \tilde{\eta}) = \nabla_{\tilde{\xi}} d\rho(\tilde{\eta}) + \rho P^\rho(\tilde{\xi}, \tilde{\eta}) + \mathcal{O}(\rho^2).$$

Taking this into account, we see that applying $\frac{-d\rho}{\rho}$ to (2.11) and ignoring terms that are $\mathcal{O}(\rho)$, we obtain

$$-\frac{1}{\rho} \nabla_{\tilde{\xi}}^\rho d\rho(\tilde{\eta}) - P^\rho(\tilde{\xi}, \tilde{\eta}) + \frac{1}{\rho^2} d\rho(\tilde{\xi}) d\rho(\tilde{\eta}),$$

for the right hand side. The first and last terms here add up to $-\nabla_{\tilde{\xi}}^\rho \frac{d\rho(\tilde{\eta})}{\rho}$, so the boundary value of the whole expression is $\nabla_{\tilde{\xi}} \tau - P^\rho(\tilde{\xi}, \tilde{\eta})$. This matches the second component of the formula for the tractor connection in [4], apart from the fact that P^ρ is the boundary value of the Schouten tensor of the connection ∇^ρ on $\overline{M}$ rather than the Schouten tensor of the induced connection on the boundary, so it remains to show that these agree.

Since the boundary is totally geodesic, the curvature of the restriction of ∇^ρ to ∂M is obtained by inserting vector fields in $\mathfrak{X}_\partial(\overline{M})$ into the curvature of ∇^ρ and then taking boundary values. However, passing to the Ricci curvature, one has to take traces over different spaces. Fixing ρ , we choose a local vector field ν such that $d\rho(\nu) \equiv 1$. Then we choose a local frame for $T\partial M$ along ∂M and extend the elements to vector fields which insert trivially into $d\rho$ to obtain a local smooth frame for $T\overline{M}$. This has the property that the dual coframe consists of $d\rho$ and of forms that restrict to the dual coframe of the initial frame for $T\partial M$. For vector fields $\tilde{\xi}, \tilde{\zeta} \in \mathfrak{X}_\partial(\overline{M})$ we denote by $\xi, \zeta \in \mathfrak{X}(\partial M)$ their restrictions to ∂M . Denoting by Ric_∂^ρ the Ricci curvature of the boundary connection and by Ric^ρ the boundary value of the Ricci curvature of the interior connection, the difference $\text{Ric}^\rho(\xi, \zeta) - \text{Ric}_\partial^\rho(\xi, \zeta)$ clearly is the boundary value of $d\rho(R(\nu, \tilde{\xi})(\tilde{\zeta}))$.

Now we can apply $d\rho$ to the defining equation for $R(\nu, \tilde{\xi})(\tilde{\zeta})$, then use (2.8) and drop terms that are $\mathcal{O}(\rho)$ to get

$$\nu \cdot d\rho(\nabla_{\tilde{\xi}}^\rho \tilde{\zeta}) - \tilde{\xi} \cdot d\rho(\nabla_\nu^\rho \tilde{\zeta}) - [\nu, \tilde{\xi}] \cdot d\rho(\tilde{\zeta}).$$

Next, for the first two summands, we can again use (2.8), but have to carry out the differentiation before dropping $\mathcal{O}(\rho)$ -terms. Up to terms that are obviously $\mathcal{O}(\rho)$, this gives

$$\begin{aligned} \nu \cdot \tilde{\xi} \cdot d\rho(\tilde{\zeta}) + d\rho(\nu)(P^\rho(\tilde{\xi}, \tilde{\zeta}) - P(\tilde{\xi}, \tilde{\zeta})) \\ - \tilde{\xi} \cdot \nu \cdot d\rho(\tilde{\zeta}) - d\rho(\tilde{\xi})(P^\rho(\nu, \tilde{\zeta}) - P(\nu, \tilde{\zeta})) - [\nu, \tilde{\xi}] \cdot d\rho(\tilde{\zeta}). \end{aligned}$$

Of course the three derivative terms cancel by definition of the Lie bracket, and $d\rho(\tilde{\xi})$ is $\mathcal{O}(\rho)$, so this part can be dropped, too. Since $d\rho(\nu) \equiv 1$ and by our assumption $\mathbf{P}(\tilde{\xi}, \tilde{\zeta})$ vanishes along the boundary, we conclude that

$$\mathrm{Ric}^\rho(\xi, \zeta) - \mathrm{Ric}_\partial^\rho(\xi, \zeta) = \mathbf{P}^\rho(\xi, \zeta) = \frac{1}{n} \mathrm{Ric}^\rho(\xi, \zeta)$$

and hence $\mathrm{Ric}_\partial^\rho(\xi, \zeta) = \frac{n-1}{n} \mathrm{Ric}^\rho(\xi, \zeta)$ which exactly implies that the resulting Schouten tensors agree. $\square$

2.10. Relation to interior tractors. While considerations about tractors inspired many of the developments in this article, it turned out that technically the weighted b -tangent bundle offers a much simpler approach that leads to stronger results. Thus we will keep the discussion of tractors rather short, but we believe that the relation of tractors to the b -tangent bundle is of independent interest.

For $\overline{M} = M \cup \partial M$, a projectively pre-compact affine connection ∇ on M defines a projective structure on M , which makes tools like tractor bundles and tractor connections available. As mentioned in Section 2.3, one cannot expect that this projective structure admits a smooth extension to all of $\overline{M}$. However, we have already seen there that some of these tools *do* extend to the boundary. In particular, the standard cotractor bundle can be defined as $J^1\mathcal{E}(1)$ on all of $\overline{M}$ and the jet exact sequence shows that it has the right composition structure on all of $\overline{M}$. Moreover, by Lemma 2.4, for the linear connection ∇^ρ on TM associated to a defining function ρ , the induced connection on $\mathcal{E}(1)$ extends smoothly to the boundary. Since such a connection splits the jet exact sequence, also the splittings of the standard cotractor bundle coming from a defining function are smooth up to the boundary.

Since general tractor bundles can be obtained by tensorial constructions from the standard cotractor bundle, all these bundles and their composition series and splittings extend to all of $\overline{M}$. In particular, we obtain an analog of the standard tractor bundle on all of $\overline{M}$ as $\mathcal{T}\overline{M} := (J^1\mathcal{E}(1))^*$ and this comes with a short exact sequence

$$0 \rightarrow \mathcal{E}(-1) \rightarrow \mathcal{T}\overline{M} \rightarrow T\overline{M}(-1) \rightarrow 0$$

and a splitting of this sequence associated to any defining function ρ . In particular, any section of $\mathcal{T}\overline{M}$ projects to a weighted vector field of weight -1 . Hence we can define a subsheaf $\Gamma_\partial(\mathcal{T}\overline{M}) \subset \Gamma(\mathcal{T}\overline{M})$ as by requiring that this projection is tangent to ∂M along ∂M . As for the tangent bundle, this gives rise to a b -tractor bundle $\mathcal{T}^b\overline{M}$, which comes with an exact sequence as above, but with $T^b\overline{M}(-1)$ replacing $T\overline{M}(-1)$.

Next, we can consider the formula for the standard tractor connection $\nabla^\mathcal{T}$ in the splittings determined by a defining function, which is mentioned in the proof of Theorem 2.13, see [4] for details. Apart from tensorial operations, this only involves the connection ∇^ρ and its Schouten tensor $\mathbf{P}^\rho$. Assuming that ∇ has totally geodesic boundary, we can apply this formula over M , to conclude that for $\xi \in \mathfrak{X}(\overline{M})$ and $s \in \Gamma(\mathcal{T}\overline{M})$, the section $\nabla_\xi^\mathcal{T} s$ of $\mathcal{T}\overline{M}|_M$ admits a smooth extension to the boundary if $\xi \in \mathfrak{X}_\partial(\overline{M})$ or $s \in \Gamma(\mathcal{T}^b\overline{M})$. Hence the tractor connection restricts to well defined operations on $\mathfrak{X}_\partial(\overline{M}) \times \Gamma(\mathcal{T}\overline{M})$ and $\mathfrak{X}(\overline{M}) \times \Gamma(\mathcal{T}^b\overline{M})$.

However, for the second operation, the values do not lie in $\Gamma(\mathcal{T}^b\overline{M})$, so it seems that one does *not* get a connection on $\mathcal{T}^b\overline{M}$.

As we have seen in Proposition 2.3, a nonzero section $\sigma \in \Gamma(\mathcal{E}(1)|_M)$ that is parallel for ∇ extends by zero to a defining density for ∂M and hence determines a global, nowhere vanishing section $I = j^1\sigma$ of $J^1(\mathcal{E}(1))$. This spans a smooth line subbundle in $J^1(\mathcal{E})$, whose annihilator is a smooth hyperplane bundle $I^\perp \subset \mathcal{T}\overline{M}$. Since σ is a defining density, it follows readily that any section of $I^\perp$ projects onto a weighted vector field that is tangent to ∂M along ∂M . This shows that we can actually view $I^\perp$ as a subbundle of $\mathcal{T}^b\overline{M}$.

The applications of tractors in [9] (where the projective structure admits a smooth extension to the boundary), which are relevant here, require that I is parallel for the tractor connection. This is equivalent to ∇ being Ricci flat, so in particular, the boundary is strongly totally geodesic. Since I is parallel, also the subbundle $I^\perp$ is parallel and hence inherits a connection, and it turns out that $I^\perp$ and this connection restrict to the projective standard tractor bundle on ∂M and its normal tractor connection. This approach can be generalized directly to the projectively pre-compact case via the operator $\mathfrak{X}_\partial(\overline{M}) \times \Gamma(J^1\mathcal{E}(1)) \rightarrow \Gamma(J^1\mathcal{E}(1))$ obtained by dualizing the corresponding operation on $\mathcal{T}\overline{M}$.

However, this can also be easily deduced from Theorem 2.13 (which works under much weaker assumptions) as follows. One easily verifies that the projection $\mathcal{T}^b\overline{M} \rightarrow T^b\overline{M}(-1)$ restricts to an isomorphism $I^\perp \rightarrow T^b\overline{M}(-1)$, which over M is just the inverse of the splitting induced by the connection ∇ . Moreover, if ∇ is Ricci-flat then, in this splitting, the normal tractor connection is simply given by ∇ , which completes the argument.

3. PROJECTIVELY PRE-COMPACT METRICS

We now turn to the case of projectively pre-compact pseudo-Riemannian metrics, which is of interest for applications to general relativity. The main topic here is the relation of projective pre-compactness to asymptotic forms of the metric. One of these forms is only available locally around boundary points in a dense open subset and plays an important role in general relativity. The second asymptotic form is available everywhere and to our knowledge is new.

3.1. An asymptotic form. In our usual setting $\overline{M} = M \cup \partial M$, we want to study a certain asymptotic form for a pseudo-Riemannian metric g on M . The basic setup is that for some function C , with properties to be specified later, and an appropriately chosen local defining function ρ , we assume that

$$(3.1) \quad h := \rho^2(g - \frac{C}{\rho^2}d\rho^2)$$

admits a smooth extension to the boundary such that the restriction of the boundary value to $T\partial M \times T\partial M$ is non-degenerate. This means that then g has the asymptotic form

$$(3.2) \quad g = \frac{C}{\rho^4}d\rho^2 + \frac{h}{\rho^2}.$$

Observe that such a form *does* depend on the choice of the defining function ρ . For $\rho = e^{-f}\hat{\rho}$ we get $d\rho = e^{-f}(d\hat{\rho} - \rho df)$ and inserting this, we see that we will get terms involving the symmetric product of $d\hat{\rho}$ and df , which are $\mathcal{O}(\rho^{-3})$ and cannot be absorbed into C unless $df = d\hat{\rho} + \mathcal{O}(\rho)$. On the other hand, a constant rescaling of ρ leads to a constant rescaling of h and C , so one may absorb constant multiples of C into a change of defining function.

The basic assumption will be that C equals a non-zero constant along the boundary, and this constant can be assumed to be ± 1 . Moreover, it has been shown in [9] that if $C = \pm 1 + \mathcal{O}(\rho^2)$ and the asymptotic form (3.2) is available locally around each boundary point, then g is projectively compact of order 1. It has also been shown in [9] that for any pseudo-Riemannian metric g which is projectively compact of order 1, such an asymptotic form is available locally around appropriate parts of the boundary, which form a dense open subset of the boundary. For applications in general relativity it turns out to be of fundamental importance to allow that $C = \pm 1 + \mathcal{O}(\rho)$ in order to allow for a non-trivial mass aspect, see [3]. We now show that such weaker fall-off conditions on the non-constant part of C imply projective pre-compactness.

Theorem 3.1. *For $\overline{M} = M \cup \partial M$, suppose that g is a pseudo-Riemannian metric on M such that locally around each $x \in \partial M$, there is a defining function ρ and a nowhere vanishing function C such that the section h defined in (3.1) admits a smooth extension to the boundary, whose boundary value is non-degenerate on $T\partial M \times T\partial M$. Suppose further that the function C has the property that for each vector field ζ that is smooth up to the boundary and such that $d\rho(\zeta) \equiv 0$, we have $\zeta \cdot C = \mathcal{O}(\rho)$. Then g is projectively pre-compact with totally geodesic boundary. If we even have $\zeta \cdot C = \mathcal{O}(\rho^2)$ for any ζ as above, then g is projectively compact (of order 1).*

Proof. Apart from the part asserting that the boundary is totally geodesic, this follows the analogous proof of Theorem 2.6 of [9], which, in particular, proves the last statement and actually implies parts of our more general result. Nevertheless, we repeat parts of that proof to ensure that the stronger assumptions on C imposed in [9] were not used implicitly. First observe that it suffices to verify the defining conditions for projective pre-compactness for the connection ∇^ρ associated to the defining function for which the asymptotic form is available. In particular, we have the tensor field h at our disposal, and restricting to a sufficiently small neighborhood of the boundary, we may assume that h is non-degenerate on $\ker(d\rho) \times \ker(d\rho)$ everywhere. As in [9], this implies that there is an analog of a Reeb field, i.e. a vector field ζ_0 such that $d\rho(\zeta_0) \equiv 1$ and such that $h(\zeta_0, \zeta) \equiv 0$ for $\zeta \in \ker(d\rho)$.

We can extend ζ_0 to a local frame by vector fields ζ_i in $\ker(d\rho)$ and then consider the frame $\tilde{\zeta}_i$ for $\ker(d\rho)$ dual to $\{\zeta_i\}$ with respect to $h|_{\ker(d\rho)}$. For a vector field ξ , we then get $\xi = d\rho(\xi)\zeta_0 + \sum_i h(\xi, \tilde{\zeta}_i)\zeta_i$, so to prove that ξ admits a smooth extension to the boundary, it suffices to prove that the functions $d\rho(\xi)$ and $h(\xi, \zeta)$ for $\zeta \in \ker(d\rho)$ admit smooth extensions.

Now the asymptotic form (3.2) readily implies that on M , we get

$$g(\eta, \zeta_0) = d\rho(\eta) \left(\frac{C}{\rho^4} + \frac{h(\zeta_0, \zeta_0)}{\rho^2} \right),$$

so to show that $d\rho(\xi)$ admits a smooth extension, it suffices to prove that $\rho^4 g(\xi, \zeta_0)$ is smooth up to the boundary. Likewise, if ζ lies in $\ker(d\rho)$, the asymptotic form shows that in the interior $g(\xi, \zeta) = \frac{1}{\rho^2} h(\xi, \zeta)$, so to show that $h(\xi, \zeta)$ is smooth up to the boundary, it suffices to prove that $\rho^2 g(\xi, \zeta)$ has this property.

To apply this to a vector field of the form $\nabla_\xi^\rho \eta$, one uses the Koszul formula to express $\nabla_\xi \eta$ to conclude that, on M , $2g(\nabla_\xi^\rho \eta, \zeta)$ can, for any vector field ζ , be written as

$$(3.3) \quad \begin{aligned} & \xi \cdot g(\eta, \zeta) - \zeta \cdot g(\xi, \eta) + \eta \cdot g(\xi, \zeta) \\ & + g([\xi, \eta], \zeta) - g([\xi, \zeta], \eta) - g([\eta, \zeta], \xi) \\ & + \frac{2}{\rho} d\rho(\xi) g(\eta, \zeta) + \frac{2}{\rho} d\rho(\eta) g(\xi, \zeta). \end{aligned}$$

Now the asymptotic form for g readily implies that we can write $\xi \cdot g(\eta, \zeta)$ as

$$-4\rho^{-5} C d\rho(\xi) d\rho(\eta) d\rho(\zeta) + \rho^{-4} \xi \cdot (C d\rho(\eta) d\rho(\zeta)) + \mathcal{O}(\rho^{-3}).$$

Putting $\zeta = \zeta_0$ and applying this to all three terms in the first line in (3.3), the ρ^{-5} -terms cancel with the last line of (3.3) and we can write (3.3) in this case as

$$(3.4) \quad \begin{aligned} & \rho^{-4} (\xi \cdot (C d\rho(\eta)) - \zeta_0 \cdot (C d\rho(\xi) d\rho(\eta) + \eta \cdot (C d\rho(\xi)))) + \\ & + g([\xi, \eta], \zeta) - g([\xi, \zeta], \eta) - g([\eta, \zeta], \xi) + \mathcal{O}(\rho^{-3}). \end{aligned}$$

Obviously, all terms in this formula are $\mathcal{O}(\rho^{-4})$, so we conclude $d\rho(\nabla_\xi^\rho \eta)$ always admits a smooth extension to the boundary (regardless of any fall off conditions on derivatives of C).

Assuming, in addition, that $\xi, \eta \in \mathfrak{X}_\partial(\overline{M})$, we have $C d\rho(\eta) = \mathcal{O}(\rho)$ and this remains true after differentiation by ξ and similarly for the last term in the first line of (3.4). Likewise, $C d\rho(\xi) d\rho(\eta) = \mathcal{O}(\rho^2)$ and differentiating in the direction ζ_0 , the result is $\mathcal{O}(\rho)$, so the whole first line of (3.4) is $\mathcal{O}(\rho^{-3})$ in this case. For $\xi, \eta \in \mathfrak{X}_\partial(\overline{M})$, we get $[\xi, \eta] \in \mathfrak{X}_\partial(\overline{M})$ and so, in each of the terms in the second line of (3.4), one of the vector fields that is inserted into g lies in $X_\partial(\overline{M})$. By the asymptotic form of g , all these terms thus also are $\mathcal{O}(\rho^{-3})$, so for $\xi, \eta \in \mathfrak{X}_\partial(\overline{M})$ we see that $d\rho(\nabla_\xi^\rho \eta)$ is $\mathcal{O}(\rho)$, so the boundary is totally geodesic.

We next analyze (3.3) in the case that $d\rho(\zeta) \equiv 0$. Then $g(\eta, \zeta) = \frac{1}{\rho^2} h(\eta, \zeta)$ for any $\zeta \in \mathfrak{X}(\overline{M})$ and differentiating this in the direction ξ , we obtain

$$-2\rho^{-3} d\rho(\xi) h(\eta, \zeta) + \mathcal{O}(\rho^{-2}).$$

Applying this to the first and the last term in the first line of (3.3), there is again a cancellation with the terms in the last line, while the first summand in the second line is $\mathcal{O}(\rho^{-2})$. Overall, we conclude that, for the case of $d\rho(\zeta) \equiv 0$, (3.3) can be written as

$$(3.5) \quad -\zeta \cdot g(\xi, \eta) - g([\xi, \zeta], \eta) - g([\eta, \zeta], \xi) + \mathcal{O}(\rho^{-2}).$$

Now $g([\xi, \zeta], \eta) = \frac{C}{\rho^4} d\rho([\xi, \zeta])d\rho(\eta) + \mathcal{O}(\rho^{-2})$, and $dd\rho(\xi, \zeta) = 0$ and $d\rho(\zeta) = 0$ imply $d\rho([\xi, \zeta]) = -\zeta \cdot d\rho(\xi)$, and like-wise for the term with ξ and η exchanged. On the other hand, the first term in (3.5) gives

$$-\zeta \cdot (\rho^{-4} C d\rho(\xi) d\rho(\eta) + \mathcal{O}(\rho^{-2})).$$

By assumption $\zeta \cdot \rho \equiv 0$, so we may pull out the factor ρ^{-4} in the first summand, while the second summand will remain to be $\mathcal{O}(\rho^{-2})$ after differentiation by ζ . Moreover, the terms in which ζ differentiates $d\rho(\xi)$ respectively $d\rho(\eta)$ cancel with the contributions from above. Hence we conclude that if $d\rho(\zeta) \equiv 0$, then without any assumption on ξ and η , (3.5) equals

$$-\rho^{-4}(\zeta \cdot C) d\rho(\xi) d\rho(\eta) + \mathcal{O}(\rho^{-2}).$$

Now by assumption $\zeta \cdot C = \mathcal{O}(\rho)$, so this sum is always $\mathcal{O}(\rho^{-3})$ and it is $\mathcal{O}(\rho^{-2})$ if either ξ or η lies in $\mathfrak{X}_\partial(\overline{M})$. Hence $h(\rho \nabla_\xi^\rho \eta, \zeta) = \rho h(\nabla_\xi^\rho \eta, \zeta)$ is smooth up to the boundary and if either ξ or η lies in $\mathfrak{X}_\partial(\overline{M})$ then $h(\nabla_\xi^\rho \eta, \zeta)$ is smooth up to the boundary, which completes the proof that ∇^ρ is projectively pre-compact.

If we even have $\zeta \cdot C = \mathcal{O}(\rho^2)$, then the whole expression is $\mathcal{O}(\rho^{-2})$ for any ξ and η , which shows that $h(\nabla_\xi^\rho \eta, \zeta)$ is always smooth up to the boundary and hence g is projectively compact. $\square$

3.2. The boundary structure. For $\overline{M} = M \cup \partial M$ consider a pseudo-Riemannian metric g on M which admits an asymptotic form (3.2) and let Ric be the Ricci curvature of g , which is just a constant multiple of the Schouten tensor of the Levi-Civita connection ∇ of g . Then by Theorems 3.1 and 2.9 and Corollary 2.10, we know that ρRic admits a smooth extension to the boundary and $\text{Ric}(\xi, \eta)$ itself extends smoothly if both ξ and η lie in $\mathfrak{X}_\partial(\overline{M})$.

To proceed towards a converse of Theorem 3.1, we have to impose a slightly stronger assumption on the Ricci curvature of g to be able to apply the results from Section 2. Our standing assumption from now on will be the following.

Definition 3.2. For $\overline{M} = M \cup \partial M$, we call a projectively pre-compact pseudo-Riemannian metric g on M with Ricci curvature Ric *asymptotically tangentially Ricci flat* if and only if $\text{Ric}(\xi, \eta)$ admits a smooth extension to the boundary if one of the fields lies in $\mathfrak{X}_\partial(\overline{M})$ and vanishes along the boundary if both fields lie in $\mathfrak{X}_\partial(\overline{M})$.

Observe that this property implies that the boundary is strongly totally geodesic, so we can apply all the developments of Section 2 to the Levi-Civita connection ∇ of g . Our first result provides a strong additional geometric structure on the boundary, which, even in the projectively compact case, was only known under the assumption that g is Ricci-flat. Note that as a density $\sigma \in \Gamma(\mathcal{E}(1))$ that is parallel for ∇ over M , we can take the appropriate power of the volume density of g here. We call this the *volume scale*.

Theorem 3.3. For $\overline{M} = M \cup \partial M$ let g be a projectively pre-compact pseudo-Riemannian metric of signature (p, q) on M , which is asymptotically tangentially Ricci flat, and let $\sigma \in \Gamma(\mathcal{E}(1))$ denote the volume scale. Then the smooth section

$\sigma^2 g \in \Gamma(S^2 T^* M(2))$ admits a smooth extension to a section of $S^2(T^b \overline{M})^*(2)$ over all of $\overline{M}$.

Moreover, the boundary value of this extension is parallel for the connection induced by ∇ and non-degenerate as a bundle metric on $T^b \overline{M}(-1)|_{\partial M}$. Hence it provides a reduction of the projective holonomy of the projective structure on ∂M obtained in Proposition 2.6 to the group $SO(p, q) \subset SL(n+1, \mathbb{R})$.

Proof. Since $g \in \Gamma(S^2 T^* M)$ and $\sigma \in \Gamma(\mathcal{E}(1))$ are parallel for the connection induced by ∇ , so is $\sigma^2 g \in \Gamma(S^2 T^* M(2))$. But we can view this as a section of $S^2(T^b \overline{M})^*(2)$ defined over M . By Theorem 2.12, ∇ smoothly extends to a linear connection on $T^b \overline{M}(-1)$ which in turn induces a linear connection on the symmetric square of the dual, i.e. on $S^2(T^b \overline{M})^*(2)$. Of course, $\sigma^2 g$ is parallel for this connection over M , so it extends by parallel transport to a smooth section of $S^2(T^b \overline{M})^*(2)$ over all of $\overline{M}$.

Since our connection is induced by a linear connection on $T^b \overline{M}(-1)$, the non-degeneracy of $\sigma^2 g$ as a bundle metric on $T^b M(-1)$ continues to hold for the extension. (View the section as a bundle map $T^b \overline{M}(-1) \rightarrow (T^b \overline{M}(-1))^*$. If the extension would be non-injective in some boundary point, one could parallelly transport a non-zero element in the kernel to obtain a contradiction to injectivity in the interior. Indeed the signature is also preserved.) Hence the boundary value is a non-degenerate bundle metric on $T^b \overline{M}(-1)|_{\partial M}$, which is parallel for the connection induced by ∇ . Under the identification from Theorem 2.13, this corresponds to a bundle metric on the projective standard tractor bundle which is parallel for the normal tractor connection and hence provides the claimed holonomy reduction. $\square$

Via the general theory of holonomy reductions of Cartan connections developed in [12], this has remarkable consequences. The case at hand has been discussed in detail in the literature, see Section 3.1 of [12], Section 3.2 of [11] and Section 3.5 of [9]. The holonomy reduction decomposes ∂M into a disjoint union as $\partial M = \partial M_+ \sqcup \partial M_0 \sqcup \partial M_-$ of so-called *curved orbits*. Here $\partial M_\pm$ are open subsets of ∂M , while ∂M_0 (if non-empty) is an embedded hypersurface which separates these two open subsets. On $\partial M_\pm$ the holonomy reduction determines pseudo-Riemannian Einstein metrics of signature $(p-1, q)$ and $(p, q-1)$ and positive and negative scalar curvature, respectively, whose Levi-Civita connections lie in (the restrictions of) the projective class. On ∂M_0 , one obtains a conformal structure of signature $(p-1, q-1)$ for which the restriction of the tractor bundle provides the conformal standard tractor bundle. Making the decomposition into curved orbits explicit in our situation is easy.

Corollary 3.4. *For $\overline{M} = M \cup \partial M$, let g be a pseudo-Riemannian metric on M , which is projectively pre-compact and asymptotically tangentially Ricci-flat. For a defining function ρ , choose a vector field ν such that $d\rho(\nu)|_{\partial M} \equiv 1$. Then the 2-density $\rho^2 \sigma^2 g(\nu, \nu)$ defined over M admits a smooth extension to the boundary.*

The boundary value $\tilde{C}$ of this density is a smooth section of the oriented line bundle $\mathcal{E}(2) \rightarrow \partial M$, which depends only on g and not on the choice of ρ . Moreover,

the decomposition $\partial M = \partial M_+ \sqcup \partial M_0 \sqcup \partial M_-$ is according to the strict sign of $\tilde{C}$ so in particular, ∂M_0 is the zero locus for $\tilde{C}$. Finally, $\tilde{C}$ is a defining density for ∂M_0 .

Proof. As we have seen in Section 2.7, the boundary value of $\rho\nu \in \Gamma(T^b\overline{M})$ is independent of the choice of ρ and spans the kernel of the projection $T^b\overline{M}|_{\partial M} \rightarrow T\partial M$. It also defines the inclusion $\mathcal{E}(-1)|_{\partial M} \rightarrow T^b\overline{M}(-1)|_{\partial M}$ which corresponds to the inclusion of the canonical line bundle into the standard tractor bundle, see Theorem 2.13. Hence the density $\tilde{C}$ exactly corresponds to the restriction of the tractor metric to $S^2\mathcal{E}(-1) \subset S^2T\partial M$. Now the result follows from Theorem 3.1 of [12]. $\square$

3.3. Necessity of asymptotic forms. Now we are ready to prove that projectively pre-compact metrics which are asymptotically tangentially Ricci flat always admit specific asymptotic forms. We first provide an asymptotic form that is valid locally around any boundary point and for any defining function for the boundary. This is new even in the case of Ricci-flat metrics that are projectively compact of order one. We also derive a relation between the boundary values of the coefficients in the expansion. Finally, we prove that locally around boundary points that lie in $\partial M_\pm$, one can choose specific defining functions for which the asymptotic form specializes to (3.2). To formulate this, we need a notation for the symmetric product of one-forms, so we put $(\varphi_1 \odot \varphi_2)(\xi, \eta) := \frac{1}{2}(\varphi_1(\xi)\varphi_2(\eta) + \varphi_1(\eta)\varphi_2(\xi))$.

Theorem 3.5. *For $\overline{M} = M \cup \partial M$, let g be a pseudo-Riemannian metric of signature (p, q) on M which is projectively pre-compact and asymptotically tangentially Ricci flat, and let ρ be any defining function for ∂M .*

(1) *There is a smooth function $C : \overline{M} \rightarrow \mathbb{R}$, a one-form $\lambda \in \Omega^1(\overline{M})$ and a section $h \in \Gamma(S^2T^*\overline{M})$ such that, over M , we have*

$$(3.6) \quad g = \frac{Cd\rho^2}{\rho^4} + \frac{\lambda \odot d\rho}{\rho^3} + \frac{h}{\rho^2}.$$

(2) *Given any asymptotic form as in (3.6), let $\underline{C}$ be the boundary value of C , and let $\underline{\lambda} \in \Omega^1(\partial M)$ and $\underline{h} \in \Gamma(S^2T^*\partial M)$ the restrictions of the boundary values of λ and h to tangential entries. Then $\underline{C}$ is a defining function for ∂M_0 and the symmetric bilinear form on the bundle $T\partial M \oplus \mathbb{R}$, over ∂M , defined by*

$$(3.7) \quad ((\xi, a), (\eta, b)) \mapsto \underline{C}ab - \frac{1}{2}a\underline{\lambda}(\eta) - \frac{1}{2}b\underline{\lambda}(\xi) + \underline{h}(\xi, \eta)$$

is non degenerate of signature (p, q) . Moreover, denoting by ∇^ρ the induced connection on $T\partial M$ and by P^ρ its Schouten tensor, we get for $\xi, \eta, \zeta \in \mathfrak{X}(\partial M)$

$$\begin{aligned} \underline{\lambda} &= -dC & \underline{h}(\xi, \eta) &= -\frac{1}{2}(\nabla_\xi^\rho \underline{\lambda})(\eta) + \underline{C}P^\rho(\xi, \eta) \\ (\nabla_\xi^\rho \underline{h})(\eta, \zeta) &= \frac{1}{2}(P^\rho(\xi, \eta)\underline{\lambda}(\zeta) + P^\rho(\xi, \zeta)\underline{\lambda}(\eta)). \end{aligned}$$

Proof. We treat (1) first. Fix a vector field ν with $d\rho(\nu) \equiv 1$ and take the density $\tilde{C} = \rho^2\sigma^2g(\nu, \nu)$ from Corollary 3.4, which extends smoothly to all of $\overline{M}$. Since $\frac{d\rho}{\rho} \in \Gamma((T^b\overline{M})^*)$, we see that $\frac{\tilde{C}}{\rho^2}d\rho^2 \in \Gamma(S^2(T^b\overline{M})^*(2))$, so also $k := \sigma^2g - \frac{\tilde{C}}{\rho^2}d\rho^2$ is a smooth section of this bundle. By construction we get $k(\rho\nu, \rho\nu) \equiv 0$. Putting

$\tilde{\lambda}(\xi) := 2k(\xi, \rho\nu)$ for $\xi \in \mathfrak{X}_\partial(\overline{M})$ defines a smooth section $\tilde{\lambda} \in \Gamma((T^b\overline{M})^*(2))$ such that $\tilde{\lambda}(\rho\nu) \equiv 0$. Hence from Section 2.7 we know that $\tilde{\lambda}$ actually lies in the subspace $\Gamma(T^*\overline{M}(2))$. Then, $\frac{\tilde{\lambda} \odot d\rho}{\rho}$ defines a smooth section of $S^2(T^b\overline{M})^*(2)$, hence also

$$(3.8) \quad \tilde{h} := k - \frac{\tilde{\lambda} \odot d\rho}{\rho} = \sigma^2 g - \frac{\tilde{C}}{\rho^2} d\rho^2 - \frac{1}{\rho} \tilde{\lambda} \odot d\rho$$

is a smooth section of $S^2(T^b\overline{M})^*(2)$. The first (defining) equation shows that for $\tilde{\xi} \in \mathfrak{X}_\partial(\overline{M})$, we get $\tilde{h}(\xi, \rho\nu) = 0$ and using Section 2.8 we conclude that $\tilde{h} \in \Gamma(S^2 T^* \overline{M}(2))$.

Putting $\sigma^\rho = \frac{\sigma}{\rho}$, we obtain a section of $\mathcal{E}(1)$ which is nowhere vanishing and we define $C := \frac{1}{(\sigma^\rho)^2} \tilde{C}$, $\lambda = \frac{1}{(\sigma^\rho)^2} \tilde{\lambda}$ and $h := \frac{1}{(\sigma^\rho)^2} \tilde{h}$. Of course, $C \in C^\infty(\overline{M}, \mathbb{R})$, $\lambda \in \Omega^1(\overline{M})$ and $h \in \Gamma(S^2 T^* \overline{M})$ and multiplying (3.8) by $\frac{1}{\sigma^2} = \frac{1}{\rho^2(\sigma^\rho)^2}$, we exactly get (3.6).

For part (2), we suppose now that we have any expression of the form (3.6), say with coefficients $\hat{C}$, $\hat{\lambda}$ and $\hat{h}$ and multiply it by $\sigma^2 = \rho^2(\sigma^\rho)^2$. Inserting two copies of $\rho\nu$ into the result, the last two terms are $\mathcal{O}(\rho)$ and $\mathcal{O}(\rho^2)$, respectively. This shows that $(\sigma^\rho)^2 \hat{C} + \mathcal{O}(\rho) = \tilde{C}$ and hence $\hat{C} = C + \mathcal{O}(\rho)$. In particular, the boundary value of $\hat{C}$ equals $\underline{C}$. This also shows that $\frac{(\sigma^\rho)^2}{\rho} \hat{\lambda} \odot d\rho + (\sigma^\rho)^2 \hat{h} = k + \mathcal{O}(\rho)$. Inserting $\rho\nu$ and $\xi \in \mathfrak{X}_\partial(\overline{M})$ into this, we conclude that $\hat{\lambda}(\xi) + \mathcal{O}(\rho) = \lambda(\xi)$, so the boundary values of $\hat{\lambda}$ and λ agree on vectors tangent to the boundary. In the same way, one concludes that the boundary values of $\hat{h}$ and h agree if both their entries are tangent to the boundary.

Hence it suffices to verify the remaining properties for the asymptotic form we have constructed. Since $C = \frac{1}{(\sigma^\rho)^2} \tilde{C}$, and σ^ρ is nowhere vanishing, Corollary 3.4 readily implies that $\underline{C}$ is a defining function for ∂M_0 . For the form we have constructed, (3.7) simply encodes the boundary value of $\frac{\sigma^2}{(\sigma^\rho)^2} g$ on $T^b\overline{M}|_{\partial M}$ in the splitting determined by ρ , so non-degeneracy follows readily.

To prove the relations between the coefficients in the asymptotic form, we work in more generality than needed here for later use. Putting $\tau^\rho := \frac{1}{\sigma^\rho} \in \mathcal{E}(-1)$, we observe that $C = \tilde{C}(\tau^\rho)^2 = \sigma^2 g(\tau^\rho \rho\nu, \tau^\rho \rho\nu)$ and the right hand side is a contraction between a smooth section of $S^2(T^b\overline{M})^*(2)$ and two sections of $T^b\overline{M}(-1)$. Hence we may differentiate the right hand side using the connection ∇ , which admits a smooth extension to the boundary, and use that $\sigma^2 g$ is parallel for this connection. Hence for any vector field $\tilde{\xi} \in \mathfrak{X}(\overline{M})$, we get

$$\tilde{\xi} \cdot C = 2\sigma^2 g(\tau^\rho \rho\nu, \nabla_{\tilde{\xi}} \tau^\rho \rho\nu).$$

Now over M , we can write

$$(3.9) \quad \nabla_{\tilde{\xi}} \tau^\rho \rho\nu = \nabla_{\tilde{\xi}}^\rho \tau^\rho \rho\nu - \tau^\rho \tilde{\xi} = \tau^\rho (\rho \nabla_{\tilde{\xi}}^\rho \nu + d\rho(\tilde{\xi})\nu - \tilde{\xi}),$$

where we used that $\nabla^\rho \tau^\rho = 0$, and observe that the sum of the last two terms inserts trivially in $d\rho$. Plugging this into the above formula for $\tilde{\xi} \cdot C$ and using that ν

inserts trivially into λ and h , we see that

$$(3.10) \quad \tilde{\xi} \cdot C = 2Cd\rho(\nabla_{\tilde{\xi}}^{\rho}\nu) + \lambda(\rho\nabla_{\tilde{\xi}}^{\rho}\nu) - \lambda(\tilde{\xi}).$$

Observe that all terms in the right hand side indeed admit a smooth extension to the boundary. Now given a vector field $\xi \in \mathfrak{X}(\partial M)$, we can extend to $\tilde{\xi} \in \mathfrak{X}_{\partial}(\overline{M})$ such that $d\rho(\tilde{\xi}) \equiv 0$ and then obtain $\xi \cdot \underline{C}$ as the boundary value of $\tilde{\xi} \cdot C$. But in this case, $\nabla_{\tilde{\xi}}^{\rho}\nu$ is smooth up to the boundary, so the middle term in the right hand side of (3.10) is $\mathcal{O}(\rho)$. Moreover, $\rho(\nu)$ is constant and $P(\tilde{\xi}, \nu)$ and (by Lemma 2.8) $P^{\rho}(\tilde{\xi}, \nu)$ admit a smooth extension to the boundary, so (2.8) shows that the first term in the right hand side is $\mathcal{O}(\rho)$, too. Hence we get $\xi \cdot \underline{C} = -\underline{\lambda}(\xi)$, which shows that $\underline{\lambda} = -d\underline{C}$.

For the second identity, we start from a vector field $\eta \in \mathfrak{X}(\partial M)$ and extend it smoothly to $\tilde{\eta} \in \mathfrak{X}_{\partial}(\overline{M})$ such that $d\rho(\tilde{\eta}) \equiv 0$. Together with the fact that ν inserts trivially into λ and h , this shows that $\sigma^2 g(\tau^{\rho}\rho\nu, \tau^{\rho}\tilde{\eta}) = \frac{1}{2}\lambda(\tilde{\eta})$. As above, we conclude that for any $\tilde{\xi} \in \mathfrak{X}(\overline{M})$ we get

$$\tilde{\xi} \cdot \lambda(\tilde{\eta}) = 2\sigma^2 g(\nabla_{\tilde{\xi}}\tau^{\rho}\rho\nu, \tau^{\rho}\tilde{\eta}) + 2\sigma^2 g(\tau^{\rho}\rho\nu, \nabla_{\tilde{\xi}}\tau^{\rho}\tilde{\eta}).$$

Now in the first term we can insert (3.9) while in the second term $d\rho(\tilde{\eta}) \equiv 0$ implies we can replace ∇ by ∇^{ρ} and then take τ^{ρ} out of the derivative. This shows that

$$(3.11) \quad \tilde{\xi} \cdot \lambda(\tilde{\eta}) = d\rho(\nabla_{\tilde{\xi}}^{\rho}\nu)\lambda(\tilde{\eta}) + 2h(\rho\nabla_{\tilde{\xi}}^{\rho}\nu, \tilde{\eta}) - 2h(\tilde{\xi}, \tilde{\eta}) + 2\frac{C}{\rho}d\rho(\nabla_{\tilde{\xi}}^{\rho}\tilde{\eta}) + \lambda(\nabla_{\tilde{\xi}}^{\rho}\tilde{\eta}).$$

By (2.8), since $\tilde{\eta} \in \mathfrak{X}_{\partial}(\overline{M})$, and $d\rho(\tilde{\eta}) \equiv 0$, $d\rho(\nabla_{\tilde{\xi}}^{\rho}\tilde{\eta}) = \mathcal{O}(\rho)$ so all terms in the right hand side admit a smooth extension to the boundary.

As before, we need the formula here only in the case that $\tilde{\xi} \in \mathfrak{X}_{\partial}(\overline{M})$ is an extension of $\xi \in \mathfrak{X}(\partial M)$ such that $d\rho(\tilde{\xi}) \equiv 0$. In this case, the first term in the right hand side of (3.11) is $\mathcal{O}(\rho)$ by (2.8), and the second term is $\mathcal{O}(\rho)$ since $\nabla_{\tilde{\xi}}^{\rho}\nu$ admits a smooth extension to the boundary. Finally, (2.8) together with $d\rho(\tilde{\eta}) \equiv 0$ and the fact that $P(\tilde{\xi}, \tilde{\eta}) = \mathcal{O}(\rho)$ implies that $d\rho(\nabla_{\tilde{\xi}}^{\rho}\tilde{\eta}) = \rho P^{\rho}(\tilde{\xi}, \tilde{\eta}) + \mathcal{O}(\rho)$. Hence (3.11) implies that

$$h(\tilde{\xi}, \tilde{\eta}) = -\frac{1}{2}\tilde{\xi} \cdot \lambda(\eta) + \frac{1}{2}\lambda(\nabla_{\tilde{\xi}}^{\rho}\tilde{\eta}) + CP^{\rho}(\tilde{\xi}, \tilde{\eta}).$$

Passing to boundary values, we obtain the claimed equation for h (taking into account that, from the proof of Theorem 2.13, we know that the boundary value of P^{ρ} is the Schouten tensor of the boundary connection).

For the last identity, we take $\eta, \zeta \in \mathfrak{X}(\partial M)$ and extend to $\tilde{\eta}, \tilde{\zeta} \in \mathfrak{X}_{\partial}(\overline{M})$ which insert trivially into $d\rho$. Then we immediately see that $\sigma^2 g(\tau^{\rho}\tilde{\eta}, \tau^{\rho}\tilde{\zeta}) = h(\tilde{\eta}, \tilde{\zeta})$, and as before we conclude that for $\tilde{\xi} \in \mathfrak{X}(\overline{M})$, we get

$$(3.12) \quad \tilde{\xi} \cdot h(\tilde{\eta}, \tilde{\zeta}) = \sigma^2 g(\tau^{\rho}\nabla_{\tilde{\xi}}^{\rho}\tilde{\eta}, \tilde{\zeta}) + \sigma^2 g(\tau^{\rho}\tilde{\eta}, \tau^{\rho}\nabla_{\tilde{\xi}}^{\rho}\tilde{\zeta}),$$

where we replaced ∇ by ∇^{ρ} as above. Expanding the first term in the right hand side, we obtain

$$\frac{1}{2\rho}d\rho(\nabla_{\tilde{\xi}}^{\rho}\tilde{\eta})\lambda(\tilde{\zeta}) + h(\nabla_{\tilde{\xi}}^{\rho}\tilde{\eta}, \tilde{\zeta}).$$

and by (2.8), we can rewrite the first term as $\frac{1}{2}(\mathbf{P}^\rho(\tilde{\xi}, \tilde{\eta}) - \mathbf{P}(\tilde{\xi}, \tilde{\eta}))\lambda(\tilde{\zeta})$. Bringing the second term to the other side and adding the corresponding term from the other summand in the right hand side of (3.12), the left hand side becomes

$$\tilde{\xi} \cdot h(\tilde{\eta}, \tilde{\zeta}) - h(\nabla_{\tilde{\xi}}^\rho \tilde{\eta}, \tilde{\zeta}) - h(\tilde{\eta}, \nabla_{\tilde{\xi}}^\rho \tilde{\zeta}).$$

Now if we specialize to the case that $\tilde{\xi}$ is an extension of $\xi \in \mathfrak{X}(\partial M)$ which inserts trivially into $d\rho$, then the boundary value of this of course is $(\nabla_\xi^\rho \underline{h})(\eta, \zeta)$. On the other hand, this implies that $\mathbf{P}(\tilde{\xi}, \tilde{\eta}) = \mathcal{O}(\rho)$, and the last claimed identity follows. $\square$

Remark 3.6. Observe that even fixing the defining function ρ , the coefficients C , λ and h in the asymptotic form (3.6) are not uniquely determined beyond the (restricted) boundary values $\underline{C}$, $\underline{\lambda}$ and $\underline{h}$. For example, the expression remains unchanged if we replace λ by $\lambda + f d\rho + \rho \alpha$ for $f \in C^\infty(\overline{M}, \mathbb{R})$ and $\alpha \in \Omega^1(\overline{M})$ and at the same time replace C by $C - \rho f$ and h by $h - \alpha \odot d\rho$. One way to fix the coefficients, which we have used implicitly in the proof already, is to choose a vector field $\nu \in \mathfrak{X}(\overline{M})$ such that $d\rho(\nu) \equiv 1$ and then require that $\lambda(\nu) \equiv 0$ and $h(\nu, \cdot) \equiv 0$.

In particular, this can be applied to the form (3.2), where we see that the question of whether $C \mp 1$ is $\mathcal{O}(\rho)$ or $\mathcal{O}(\rho^2)$ only makes sense once requires that the cross term $\frac{1}{\rho^3} \lambda \odot d\rho$ vanishes identically. So this pins down the coefficients here.

In particular, Theorem 3.5 shows that, for projectively pre-compact metrics, the form (3.2) can only be available on open subsets whose intersection with ∂M is either contained in ∂M_+ or in ∂M_- (corresponding to $C = 1 + \mathcal{O}(\rho)$ or $-1 + \mathcal{O}(\rho)$). On such subsets, one can easily obtain the form (3.2) (for an appropriate defining function) from Theorem 3.5 using the ideas in Remark 3.6.

Corollary 3.7. *For $\overline{M} = M \cup \partial M$, let g be a pseudo-Riemannian metric on M which is projectively pre-compact and asymptotically tangentially Ricci flat.*

Then locally around each point of $\partial M_\pm$, there is a defining function $\hat{\rho}$ with respect to which g can be expressed as in (3.2) with $C = \pm 1 + \mathcal{O}(\rho)$.

Proof. Let us start from an expression (3.6) for g for a defining function ρ . Then for the defining function $\hat{\rho} = e^f \rho$ we get $d\hat{\rho} = \hat{\rho} df + e^f d\rho$ and inserting $\rho = e^{-f} \hat{\rho}$ and $d\rho = e^{-f}(d\hat{\rho} - \hat{\rho} df)$ into (3.6), we see that we get an asymptotic form (3.6) for g with respect to $\hat{\rho}$, which starts with $\frac{e^{2f} C}{\hat{\rho}^4} d\hat{\rho}^2$. Now on the subset where C is non-vanishing, we can choose $f := -\frac{1}{2} \log(|C|)$ and so we get an expression of the form

$$(3.13) \quad \frac{\pm 1}{\hat{\rho}^4} d\hat{\rho}^2 + \frac{1}{\hat{\rho}^3} (\hat{\lambda} \odot d\hat{\rho}) + \frac{1}{\hat{\rho}^2} \hat{h}.$$

Moreover, by Theorem 3.5, the boundary valued of $\hat{\lambda}$ vanishes in tangential directions, so there are $\varphi \in C^\infty(\overline{M}, \mathbb{R})$ and $\alpha \in \Omega^1(\overline{M})$ such that $\hat{\lambda} = \varphi d\hat{\rho} + \hat{\rho} \alpha$. This

means that (3.13) equals

$$\frac{\pm 1 + \rho\varphi}{\hat{\rho}^4} d\hat{\rho}^2 + \frac{1}{\hat{\rho}^2} (\hat{h} + \alpha \odot d\hat{\rho}),$$

so this has the claimed form. $\square$

3.4. Obstructions to projective compactness. We have already identified a complete obstruction to projective compactness in the general setting of connections in Section 2.3 via the boundary values of $\rho\nabla_\xi^\rho\eta$ for $\xi, \eta \in \mathfrak{X}(\overline{M})$. As we have seen there, it is of the form $d\rho(\xi)|_{\partial M}d\rho(\eta)|_{\partial M}\psi$ for a vector field $\psi \in \mathfrak{X}(\partial M)$. Now for general asymptotic forms (3.6) we can use the computations from the proof of Theorem 3.5 to deduce an equivalent characterization of projective compactness for metrics.

Theorem 3.8. *For $\overline{M} = M \cup \partial M$, let g be a projectively pre-compact pseudo-Riemannian metric and assume that for some defining function ρ and vector field $\nu \in \mathfrak{X}(\overline{M})$ such that $d\rho(\nu) \equiv 1$, we are given an expression (3.6) such that $\lambda(\nu) \equiv 0$ and $h(\nu, \cdot) \equiv 0$. (In particular the asymptotic form constructed in the proof of Theorem 3.5 has this property.)*

Then g is projectively compact if and only if

$$(3.14) \quad \nu \cdot C - 2Cd\rho(\nabla_\nu^\rho\nu) = \mathcal{O}(\rho)$$

and for any $\tilde{\zeta} \in \mathfrak{X}(\overline{M})$ such that $d\rho(\tilde{\zeta}) \equiv 0$, we get

$$(3.15) \quad -\nu \cdot \lambda(\tilde{\zeta}) + \lambda(\nabla_\nu^\rho\tilde{\zeta}) + d\rho(\nabla_\nu^\rho\nu)\lambda(\tilde{\zeta}) + 2\frac{C}{\rho}d\rho(\nabla_\nu^\rho\tilde{\zeta}) = \mathcal{O}(\rho).$$

Observe that the first two terms in (3.15) can be written as $-(\nabla_\nu^\rho\lambda)(\tilde{\zeta})$ since $\nabla_\nu^\rho\lambda$ makes sense as acting on $\mathfrak{X}_\partial(\overline{M})$. Likewise, using (2.8), we see that $d\rho(\nabla_\nu^\rho\nu) = \rho P^\rho(\nu, \nu) - \rho P(\nu, \nu)$ and $\frac{1}{\rho}d\rho(\nabla_\nu^\rho\tilde{\zeta}) = P^\rho(\nu, \tilde{\zeta}) - P(\nu, \tilde{\zeta})$. Correspondingly, we can interpret (3.14) and (3.15) as a system of “asymptotic ODE” on $C \in C^\infty(\overline{M}, \mathbb{R})$ and $\lambda \in \Gamma((T^b\overline{M})^*)$ of the form

$$\nu \cdot C = 2fC + \mathcal{O}(\rho) \quad \nabla_\nu^\rho\lambda = f\lambda + 2C\varphi + \mathcal{O}(\rho),$$

where $f := \rho P^\rho(\nu, \nu) - \rho P(\nu, \nu)$ and $\varphi(\tilde{\zeta}) = P^\rho(\nu, \tilde{\zeta}) - P(\nu, \tilde{\zeta})$.

Proof. The metric g is projectively compact if and only if the boundary value $\psi \in \Gamma(T\partial M)$ of $d\rho(\rho\nabla_\nu^\rho\nu)$ vanishes identically. By the non-degeneracy statement in Theorem 3.5 this is equivalent to vanishing of $\underline{\lambda}(\psi)$ and of $\underline{h}(\psi, \zeta)$ for any vector field $\zeta \in \mathfrak{X}(\partial M)$. Now we can use formula (3.10) from the proof of Theorem 3.5 (which is valid for any $\xi \in \mathfrak{X}(\overline{M})$) in the case $\xi = \nu$ to obtain

$$\lambda(\nabla_\nu^\rho\nu) = \nu \cdot C - 2Cd\rho(\nabla_\nu^\rho\nu),$$

which immediately shows that (3.14) is equivalent to vanishing of $\underline{\lambda}(\psi)$.

For (3.15), we extend ζ to a vector field $\tilde{\zeta} \in \mathfrak{X}(\overline{M})$ such that $d\rho(\tilde{\zeta}) \equiv 0$. Then we use formula (3.11) from the proof of Theorem 3.5 (which is valid for any $\xi \in \mathfrak{X}(\overline{M})$) in the case $\xi = \nu$ to obtain

$$-h(\rho\nabla_\nu^\rho\nu, \tilde{\zeta}) = -\frac{1}{2}\nu \cdot \lambda(\tilde{\zeta}) + \frac{1}{2}\lambda(\nabla_\nu^\rho\tilde{\zeta}) + \frac{1}{2}d\rho(\nabla_\nu^\rho\nu)\lambda(\tilde{\zeta}) + \frac{C}{\rho}d\rho(\nabla_\nu^\rho\tilde{\zeta}).$$

Passing to boundary values, (3.15) is evidently equivalent to vanishing of $\underline{h}(\psi, \zeta)$ for any $\zeta \in \mathfrak{X}(\partial M)$. $\square$

So let us pass to the question of projective compactness of a metric of the form (3.2). As originally proved in [9] and reproved in Theorem 3.1 above, $C = \pm 1 + \mathcal{O}(\rho^2)$ is sufficient for projective compactness. The asymptotic form (3.2) plays a prominent role in general relativity, going back at least to [3]. There the $\mathcal{O}(\rho)$ -term of $C \mp 1$ is referred to as the mass aspect and it plays a crucial role in the applications to GR. So in that sense, projective compactness is incompatible with mass, which has also been pointed out in [5, 6, 7]. From this point of view, it would be desirable to prove that for a metric of the form (3.2) projective compactness forces $C = \pm 1 + \mathcal{O}(\rho^2)$. Our last result shows that this is indeed true under slightly stronger assumptions on the asymptotics of the Ricci curvature of g (which should be satisfied in the cases relevant for GR which mostly deal with Ricci-flat metrics).

Theorem 3.9. *For $\overline{M} = M \cup \partial M$, let g be a projectively pre-compact pseudo-Riemannian metric with an asymptotic form (3.2). Assume further that its Ricci curvature admits a smooth extension to the boundary and the boundary value of this curvature vanishes if one of the entries is tangent to the boundary.*

Then g is projectively compact if and only if $C = \pm 1 + \mathcal{O}(\rho^2)$.

Proof. It remains to show that projective compactness of g forces $C = \pm 1 + \mathcal{O}(\rho^2)$, so we assume that g is projectively compact. We cannot directly use the computations from Theorem 3.5 here, since in the case of (3.2), we cannot say anything about $h(\nu, \cdot)$ for a vector field $\nu \in \mathfrak{X}(\overline{M})$ such that $d\rho(\nu) \equiv 1$. It would be easy to redo these computations in the current setting, but we can also modify the asymptotic form along the lines of Remark 3.6 and then use Theorem 3.8.

Starting from (3.2) for a defining function ρ , we fix $\nu \in \mathfrak{X}(\overline{M})$ such that $d\rho(\nu) \equiv 1$. Define $\varphi := h(\nu, \nu) \in C^\infty(\overline{M}, \mathbb{R})$ and for $\xi \in \mathfrak{X}(\overline{M})$ consider $\alpha(\xi) := 2(h(\nu, \xi) - d\rho(\xi)\varphi)$. This clearly defines $\alpha \in \Omega^1(\overline{M})$ such that $\alpha(\nu) \equiv 0$ and we define $\hat{h} \in \Gamma(S^2 T^* \overline{M})$ by $\hat{h} := h - \alpha \odot d\rho - \varphi d\rho^2$. Then $\hat{h}(\nu, \eta) = h(\eta, \nu) - \varphi d\rho(\eta) + \frac{1}{2}\alpha(\eta) = 0$. Thus (3.2) can be rewritten as

$$g = \frac{C + \rho^2 \varphi}{\rho^4} d\rho^2 + \frac{1}{\rho^3} (\rho \alpha) \odot d\rho + \hat{h},$$

and by construction ν inserts trivially into $\rho \alpha$ and into $\hat{h}$. By Theorem 3.8, projective compactness implies that

$$\nu \cdot (C + \rho^2 \varphi) = 2C d\rho(\nabla_\nu^\rho \nu) + \mathcal{O}(\rho),$$

and of course the left hand side equals $\nu \cdot C + \mathcal{O}(\rho)$. As we have noted after Theorem 3.5, $d\rho(\nabla_\nu^\rho \nu) = \rho P^\rho(\nu, \nu) - \rho P(\nu, \nu)$. By projective compactness ∇^ρ admits a smooth extension to the boundary and hence so does its Schouten-tensor so $\rho P^\rho(\nu, \nu) = \mathcal{O}(\rho)$. But by our stronger curvature assumptions also $P(\nu, \nu)$ admits a smooth extension to the boundary and hence $\nu \cdot C = \mathcal{O}(\rho)$. Since for $C = \pm 1 + \rho f$, we get $\nu \cdot C = f + \mathcal{O}(\rho)$, this completes the proof. $\square$

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